

Prizes CP

The United States federal government should:

- establish an innovation prize system for new technology, cyber, and quantum computing, and*
- abolish patents.*

Prizes spur innovation comparatively better than patents – higher publicity, more contestants, more follow-on innovation, and they avoid deadweight loss from monopoly pricing.

Chavez 22 — Anthony Chavez, Professor of Law at Chase College of Law, B.S. in Accounting from Loyola Marymount University, J.D. from Yale Law School, 2022 ("Turning Carbon into Gold: Incentivizing the New Alchemy," *Duke environmental Law and Policy Forum*, Available Online at <https://delpf.law.duke.edu/article/turning-carbon-into-gold-chavez-vol32-iss1/>, Accessed 08-18-2024)

Analysts have identified a number of benefits that prizes have over other incentive systems, especially patents. A primary advantage is that prizes can encourage researching technologies that markets do not value sufficiently. 254 Markets fail to encourage the development of technologies whose market prices do not reflect their value. 255 Prizes (and grants) "reward innovations that 'are publicly valued but not privately marketable.'" 256 A government prize, for instance, is essentially a government intervention into the markets where they have failed to sufficiently incentivize a public good. 257 Prizes are also [*31] well suited to encourage high-risk research on technologies that are still at their earliest stages. 258 Similarly, they might also be well suited to encouraging research in circumstances where scientific opinion is divided concerning the appropriate path to direct technological development. 259 In other words, prizes fit circumstances where goals are clear, but the best paths to achieve them are less certain. 260 A number of competitions have demonstrated this pattern. Specifically, the Longitude Bill, the Netflix Prize, and the Progressive Insurance Automotive X Prize all involved identifiable goals where the means to achieve them were uncertain. 261 By establishing criteria that encourage research in particular areas, prizes can offer ex ante incentives to pursue a targeted objective. 262 Finally, by setting criteria that suggest a particular result, organizers can signal the importance of a particular issue. 263

Prizes also avoid the deadweight loss of patents. 264 Unlike the X Prize Foundation's provisions, organizers of other prizes usually require winners of traditional prizes to place their intellectual property into the public domain. 265 Consequently, if the proceeds from prizes are comparable to the returns inventors would receive from patent rights, then inventors receive fair compensation while the public gains ready access to their inventions without incurring the deadweight loss of monopoly patent rights. 266 In programs like those of the X Prize Foundation, which allow innovators to retain their patent rights, the prizes augment the rewards of the patent system. 267 This may be especially helpful where markets do not price in the social value of innovations because the prizes can help to close the gap between investment cost and market compensation. 268

Another advantage of prizes is that they often broaden the pool of [*32] prospective innovators. Prizes attract persons who might not otherwise engage in commercial efforts 269 or who might not otherwise have pursued a particular field. 270 The competitive nature of prizes may also attract new parties interested in pursuing the awards. 271 As with the Longitude Act, prizes can attract parties who might normally not be among the anticipated pool of prospective inventors. 272 Also, by broadening the range of participants in the competition, prizes bring the advantages of diversification to the efforts, thereby increasing the likelihood that the competition will generate solutions to the problem. 273 Prizes broaden the pool of participants in part because they also bring attention to their subject matter. 274

The publicity surrounding prizes appears to be a significant advantage that they provide. Prizes can both generate interest in a topic and add to the prestige of participants. 275 Indeed, historical analysis suggests that the publicity from prizes without cash

awards can encourage innovation. 276 One result of publicity is that it attracts additional inventors to the subject. 277 Furthermore, prizes often signal to other possible inventors the significance of innovation in a particular area. 278

Unlike patents, prizes do not impede - and can support - follow-on innovations. As discussed previously, the exclusive rights and monopoly pricing of the patent system can deter follow-on innovations. 279 Empirical evidence indicating that patents deter such innovations suggests that prizes would be more favorable to them. 280 Similarly, assuming that large awards encourage substantial research efforts for breakthrough inventions, organizers can use a series of smaller prizes to encourage follow-on inventions. 281

1NC Sui Generis 1NC

*The United States federal government should **adopt a holistic and factually determined patent-eligibility test** via a sui generis intellectual property regime.*

Sui generis is a unique form of IPR external to patents - it solves the aff and the perm fails

Kazeeva 24 [Iana Kazeeva is a postdoctoral researcher at the Department of Innovation and Digitalisation in Law at the University of Vienna. She studied international commercial and tax law in Moscow and holds a Candidate of Sciences degree (Ph.D. equivalent) in international financial law (dissertation on fiscal decentralization). Prior to joining the Department, she worked at the Mountain View office of a California-based public software-as-a-service company, primarily engaged in reviewing and negotiating technology and commercial contracts. She also gained practical experience as a senior legal counsel at the legal department of a global producer of mineral fertilizers, focusing on project finance and legal compliance. At the Department, the primary focus of her research is on intellectual property, data security, and data protection law., Sui Generis Intellectual Property Protection Comparison of EU and U.S. Regulatory Perspectives in Law, Business and Innovation, ALB]

1.2 The Reasons for Sui Generis Intellectual Property

Protection

Although the legal concept of intellectual property has been developing over hundreds of years and its legal framework is well developed and includes multiple forms of legal protection, there are certain categories of subject matter that cannot be effectively protected by the existing intellectual property legal regimes. **This is the case when the need for a sui generis, i.e., of its own kind, protection emerges**. For instance, the purpose of early utility models laws was to protect functional improvements of handtools, which neither patent nor trade secret law effectively protected.²⁰ Besides filling such legal gaps, another reason for introducing sui generis regimes is derived from the utilitarian theory of intellectual property protection, namely, the need to find the balance between the public's access to inventions and creations, on the one hand, and the proprietors' interests, on the other hand

The list of subject matter, for which the need for sui generis intellectual property protection has been either claimed for, or acknowledged, in a number of jurisdictions or at international level, includes databases, designs, plant varieties, semiconductor chips, geographical indications, traditional knowledge, etc.²¹ Recently, the issue of granting sui generis protection for works generated by artificial intelligence models has also been raised by scholars and practitioners.²² This book will further discuss the forms of intellectual property protection available for databases, designs, and plant varieties, representing three different areas of creations of the human intellect: physical compilations of data and digital technologies (databases), creative ideas (designs) and plant biology (plant varieties).

Databases and other compilations can generally be protected under copyright law, which is stipulated at both national and international levels. However, databases and compilations are often referred to as “low-authorship factual works”,²³ which demonstrates the main legal problem that authors face when trying to get copyright protection for databases: lack of originality. Indeed, before the introduction of the sui generis protection for databases in the EU, only three among the fifteen EU Member States protected a significant number of databases using copyright, which was possible due to applying a very low threshold of originality.²⁴ Furthermore, the courts in the U.S.²⁵ and EU²⁶ confirmed that copyright law does not protect databases that result merely from economic investment or intellectual effort.

As to ornamental designs, which are in principle patentable in the U.S., in practice the process of getting a patent for such designs is too costly and slow for the fastmoving, short-lived products of mass consumption.²⁷ At the same time, copyright protection is in most countries strictly limited for ornamental designs. For example, in the U.S., clothing, furniture, dinnerware, etc., are considered useful articles having intrinsic utilitarian functions, and, therefore, the design of these articles cannot be protected by copyright law.²⁸

For innovations in the agricultural industry, trade secret and patent laws could be the appropriate instruments for legal protection. Use of trade secret laws for protection of genetic information in an inbred plant line was upheld in *Pioneer HiBred International v. Holden Foundation Seed*,²⁹ despite the fact that parent plants are grown outdoors over a wide area, which contradicts the traditional trade secret law requirement to keep information secret. However, trade secret protection is only available for those crops that are sold as a hybrid seed, as opposed to self-pollinated crops, because in that case the parental plant lines can be kept secret.³⁰ Patent law does not provide full relief either, since the products of nature are generally excluded from patentable subject matter in the U.S.³¹ The majority of the leading European countries are also reluctant to bring life forms within the domestic patent laws, at least for plants obtained through conventional breeding processes, since granting such patents could potentially lead to creation of monopolies and raise food prices.³² The need for sui generis protection for plant varieties has also been recognized at the international level. Particularly, Article 27.3 of the TRIPS Agreement states that “members shall provide protection for plant varieties either by patents or by an effective sui generis system, or by any combination thereof”.

Indeed, the mentioned categories of subject matter can be protected by a combination of the standard intellectual property regimes. For instance, an ornamental feature of a design can be protected by a design patent, by copyright as the author’s expression, and by trademark law (in the form of a trade dress) as a source indicator. However, the overlapping rights make it difficult for intellectual property users to determine the scope of such different rights, particularly when those rights have different terms or limitations. In order to resolve this issue, some scholars suggested that courts should require intellectual property owners to choose the right they want enforced or follow the requirements of the regime whose protection term expires first. Besides, although the courts might not object to invoking overlapping intellectual property regimes as long as the requirements for each regime are met, it is not clear to what extent a particular intellectual property right should be enforced. Therefore, such doctrinal overlap presents a temptation to intellectual property owners to push for even stronger protection of their intellectual property rights.³³

Another possible instrument of protection is the unfair competition laws. However, unfair competition laws are a completely different legal concept and are applicable only in certain situations and therefore do not provide enough protection. In particular, the law of unfair competition is only applicable in competitive situations, whereas the sui generis right also covers acts committed in non-competitive

situations. Besides, the law of unfair competition sanctions unfair behavior a posteriori, whereas the sui generis rights create rights a priori and grant a term of protection.³⁴

Sui generis solves innovation best – plants, geography, and databases prove

Wysebridge 24 [Wysebridge = company for patent bar test prep, “Sui generis: Exploring a Patent, the MPEP, and the Patent Bar,” <https://wysebridge.com/sui-generis-exploring-a-patent-the-mpep-and-the-patent-bar>, ALB]

Sui generis, a Latin term meaning “of its own kind,” is a concept that holds great importance in the world of intellectual property. In this article, we will delve into the depths of patents, the Manual of Patent Examining Procedure (MPEP), and the patent bar. By understanding these three key components, you will gain valuable insights into the intricate world of intellectual property rights.

Understanding the Concept of Sui Generis

Before we embark on our exploration, let's first grasp the definition and importance of the term sui generis. Simply put, it refers to something that is unique or in a class of its own. When it comes to intellectual property, the term often refers to rights or protections that are distinct from other forms of legal protection.

Definition and Importance of Sui Generis

In the realm of intellectual property, sui generis refers to legal frameworks specially designed to address unique types of creations or inventions. These frameworks recognize that certain creations require their own set of rules and regulations to ensure adequate protection.

The concept of sui generis is crucial as it provides a tailored approach to intellectual property rights, ensuring that innovative and creative works receive the protection they deserve. By having specific legal mechanisms, it enables creators and inventors to enjoy exclusive rights over their creations, encouraging innovation and fostering economic growth.

When it comes to intellectual property, the world is a vast landscape filled with diverse creations and inventions. While some forms of intellectual property are well-defined and fall under existing legal frameworks, others require sui generis protection.

Sui Generis in the Context of Intellectual Property

Intellectual property encompasses a wide range of creations, including inventions, designs, trademarks, and copyrights. Each of these categories has its own set of rules and regulations to ensure protection and encourage innovation. However, there are certain creations that do not fit neatly into these existing categories, calling for sui generis protection.

For example, plant varieties, geographical indications, and databases are considered sui generis within the domain of intellectual property. These creations possess unique characteristics that demand specialized protection, and therefore, they are subject to specific legal regimes tailored to their nature.

Plant varieties, for instance, require sui generis protection due to their distinct nature. The development of new plant varieties through breeding or genetic manipulation is a complex process that involves

extensive research, investment, and time. To incentivize breeders and protect their efforts, sui generis systems such as plant variety rights have been established. These systems grant breeders exclusive rights over their new plant varieties, allowing them to control their commercial exploitation and prevent unauthorized use.

Geographical indications, on the other hand, refer to indications that identify a product as originating from a specific geographical area, where a particular quality, reputation, or other characteristic is attributable to its geographical origin. These indications, like Champagne or Parmigiano-Reggiano, require sui generis protection to safeguard their reputation and prevent misuse. Sui generis systems for geographical indications establish a link between the product and its geographical origin, ensuring that only products meeting specific criteria can bear the protected indication.

Databases, in the age of information, play a crucial role in various sectors. These databases, which compile and organize large amounts of data, require sui generis protection to encourage investment in their creation and maintenance. Sui generis database rights grant the creators of databases the exclusive right to control the extraction and reutilization of their contents. This protection ensures that the efforts put into collecting and organizing data are appropriately rewarded.

By recognizing the need for sui generis protection in these specific areas, intellectual property laws adapt to the ever-evolving landscape of innovation. Sui generis frameworks allow for the creation of tailored regulations that address the unique challenges and requirements posed by these distinct creations, promoting a fair and balanced intellectual property system.

Innovation prevents extinction

Schmidt '23 [Eric; Former CEO @ Google; Foreign Affairs, “Innovation Power,” vol. 102]

Ukraine's success can be credited in part to the resolve of the Ukrainian people, the weakness of the Russian military, and the strength of Western support. But it also owes to the defining new force of international politics: innovation power. Innovation power is the ability to invent, adopt, and adapt new technologies. It contributes to both hard and soft power. High-tech weapons systems increase military might, new platforms and the standards that govern them provide economic leverage, and cutting-edge research and technologies enhance global appeal. There is a long tradition of states harnessing innovation to project power abroad, but what has changed is the self-perpetuating nature of scientific advances. Developments in artificial intelligence in particular not only unlock new areas of scientific discovery; they also speed up that very process. Artificial intelligence supercharges the ability of scientists and engineers to discover ever more powerful technologies, fostering advances in artificial intelligence itself as well as in other fields—and reshaping the world in the process.

The ability to innovate faster and better—the foundation on which military, economic, and cultural power now rest—will determine the outcome of the great-power competition between the United States and China. For now, the United States remains in the lead. But China is catching up in many areas and has already surged ahead in others. To emerge victorious from this century-defining contest, business as usual will not do. Instead, the U.S. government will have to overcome its stultified bureaucratic impulses, create favorable conditions for innovation, and invest in the tools and talent needed to kick-start the virtuous cycle of technological advancement. It needs to commit itself to promoting innovation in the service of the country and in the service of democracy. At stake is nothing less than the future of free societies, open markets, democratic government, and the broader world order.

KNOWLEDGE IS POWER

The nexus between technological innovation and global domination dates back centuries, from the muskets the conquistador Francisco Pizarro wielded to defeat the Inca Empire to the steamboats Commodore Matthew Perry commanded to force the opening of Japan. But the sheer speed at which innovation is happening has no precedent. Nowhere is this change clearer than in one of the foundational technologies of our time: artificial intelligence.

Today's AI systems can already provide key advantages in the military domain, where they are able to parse millions of inputs, identify patterns, and alert commanders to enemy activity. The Ukrainian military, for example, has used AI to efficiently scan intelligence, surveillance, and reconnaissance data from a variety of sources. Increasingly, however, AI systems will move beyond merely assisting human decision-making and start making decisions themselves. John Boyd, a military strategist and U.S. Air Force colonel, coined the term "OODA loop"—observe, orient, decide, act—to describe the decision-making process in combat. Crucially, AI will be able to execute each part of the OODA loop much faster. Conflict can happen at the speed of computers, not the speed of people. As a result, command-and-control systems that rely on human decision-makers—or, worse, complex military hierarchies—will lose out to faster, more efficient systems that team machines with humans.

In previous eras, the technologies that shaped geopolitics—from bronze to steel, steam power to nuclear fission—were largely singular. There was a clear threshold of technological mastery, and once a country reached it, the playing field was leveled. Artificial intelligence, by contrast, is generative in nature. By presenting a platform for continuous scientific and technological innovation, it can lead to yet more innovation. That phenomenon makes the AI age fundamentally different from the Bronze Age or the steel age. Rather than natural resource wealth or mastery of a given technology, the source of a country's power now lies in its ability to continuously innovate.

This virtuous cycle will only get faster and faster. Once quantum computing comes of age, superfast computers will allow for the processing of ever-larger amounts of data, producing ever-smarter AI systems. These AI systems, in turn, will be able to produce breakthrough innovations in other emerging fields, from synthetic biology to semiconductor manufacturing. Artificial intelligence will change the very nature of scientific research. Instead of making progress one study at a time, scientists will discover the answers to age-old questions by analyzing massive data sets, freeing the world's smartest minds to devote more time to developing new ideas. As a foundational technology, AI will be critical in the race for innovation power, lying behind countless future developments in drug discovery, gene therapy, material science, and clean energy—and in AI itself. Faster airplanes did not help build faster airplanes, but faster computers will help build faster computers.

Even more powerful than today's artificial intelligence is a more comprehensive technology—for now, given current computing power, still hypothetical—called "artificial general intelligence," or AGI. Whereas traditional AI is designed to solve a discrete problem, AGI should be able to perform any mental task a human can and more. Imagine an AI system that could answer seemingly intractable questions, such as the best way to teach a million children English or to treat a case of Alzheimer's disease. The advent of AGI remains years, perhaps even decades, away, but whichever country develops the technology first will have a massive advantage, since it could then use AGI to develop ever more advanced versions of AGI, gaining an edge in all other domains of science and technology in the process. A breakthrough in this field could usher in an era of predominance not unlike the short period of nuclear superiority the United States enjoyed in the late 1940s.

Whereas many of AI's most transformative effects are still far off, innovation in drones is already upending the battlefield. In 2020, Azerbaijan employed Turkish- and Israeli-made drones to gain a decisive advantage in its war against Armenia in the disputed Nagorno-Karabakh region, racking up battlefield victories after more than two decades of military stalemate. Similarly, Ukraine's fleet of drones—many of which are low-cost commercial models repurposed for reconnaissance behind enemy lines—have played a critical role in its successes.

Drones offer distinct advantages over traditional weapons: they are smaller and cheaper, offer unmatched surveillance capabilities, and reduce soldiers' risk exposure. Marines in urban warfare, for example, could be accompanied by microdrones that serve as their eyes and ears. Over time, countries will improve the hardware and software powering drones to outinnovate their rivals. Eventually, autonomous weaponized drones—not just unmanned aerial vehicles but

also ground-based ones—will replace soldiers and manned artillery altogether. Imagine an autonomous submarine that could quickly move supplies into contested waters or an autonomous truck that could find the optimal route to carry small missile launchers across rough terrain. Swarms of drones, networked and coordinated by AI, could overwhelm tank and infantry formations in the field. In the Black Sea, Ukraine has used drones to attack Russian ships and supply vessels, helping a country with a minuscule navy constrain Russia's mighty Black Sea Fleet. Ukraine offers a preview of future conflicts: wars that will be waged and won by humans and machines working together.

As developments in drones make clear, innovation power underlies military power. First and foremost, technological dominance in crucial domains bolsters a country's ability to wage war and thus strengthens its deterrent capabilities. But innovation also shapes economic power by giving states leverage over supply chains and the ability to make the rules for others. Countries reliant on natural resources or trade, especially those that must import rare or foundational goods, face vulnerabilities others do not.

Consider the power China can wield over the countries it supplies with communications hardware. It is no surprise that countries dependent on Chinese-supplied infrastructure—such as many countries in Africa, where components produced by Huawei make up about 70 percent of 4G networks—have been loath to criticize Chinese human rights violations. Taiwan's primacy in semiconductor manufacturing, likewise, provides a powerful deterrent against invasion, since China has little interest in destroying its largest source of microchips. Leverage also accrues to countries pioneering new technologies. The United States, thanks to its role in the foundation of the Internet, has for decades enjoyed a seat at the table defining Internet regulations. During the Arab Spring, for example, the fact that the United States was home to technology companies that provided the backbone of the Internet enabled those companies to refuse Arab governments' censorship requests.

Less obvious but also crucial, technological innovation buoys a country's soft power. Hollywood and tech companies such as Netflix and YouTube have built up a trove of content for an increasingly global consumer base—all the while, helping spread American values. Such streaming services project the American way of life into living rooms around the world. Similarly, the prestige associated with U.S. universities and the opportunities for wealth creation created by U.S. companies attract strivers from across the globe. In short, a country's ability to project power in the international sphere—militarily, economically, and culturally—depends on its ability to innovate faster and better than its competitors.

RACE TO THE TOP

The main reason innovation now lends such a massive advantage is that it begets more innovation. In part, it does so because of the path dependency that arises from clusters of scientists attracting, teaching, and training other great scientists at research universities and large technology companies. But it also does so because innovation builds on itself. Innovation relies on a loop of invention, adoption, and adaptation—a feedback cycle that fuels yet more innovation. If any link in the chain breaks, so, too, does a country's ability to innovate effectively.

A lead in invention is typically built on years of prior research. Consider the way the United States led the world into the 4G era of telecommunications. The rollout of 4G networks across the country facilitated the early development of mobile applications such as Uber that required faster cellular data connections. With that lead, Uber was able to refine its product in the United States so it could roll it out in developing countries. This led to many more customers—and much more feedback to incorporate—as the company adapted its product for new markets and new releases.

But the moat around countries that enjoy structural advantages in technology is shrinking. Thanks in part to more accessible academic research and the rise of open-source software, technologies now diffuse more quickly around the world. The availability of new advances has helped competitors catch up at record speed, as China eventually did in 4G. Although some of China's recent technological success stems from economic espionage and a disregard for patents, much of it traces back to innovative, rather than derivative, efforts to adapt and implement new technology.

Indeed, Chinese companies have enjoyed resounding success in adopting and commercializing foreign technological breakthroughs. In 2015, the Chinese Communist Party laid out its “Made in China 2025” strategy to achieve self-sufficiency in high-tech industries such as telecommunications and AI. As part of this bid, it announced an economic plan of “dual circulation,” whereby China intends to boost both domestic and foreign demand for its goods. Through public-private partnerships, direct subsidies to private companies, and support for state-backed companies, Beijing has poured billions of dollars into ensuring it comes out ahead in the race for technological supremacy. So far, the record is mixed. China is ahead of the United States in some technologies yet lags in others.

It is hard to say whether China will seize the lead in AI, but top officials in Beijing certainly think it will. In 2017, Beijing announced plans to become the global leader in artificial intelligence by 2030, and it may achieve that goal even earlier than expected. China has already accomplished its goal of becoming the world’s leader in AI-based surveillance technology, which it not only uses to control dissidents at home but also sells to authoritarian governments abroad. China still ranks behind the United States in attracting the best minds in AI, with almost 60 percent of top-tier researchers working in U.S. universities. But China’s loose privacy laws, mandatory data collection, and targeted government funding give the country a key advantage. Indeed, it already leads in the production of autonomous vehicles.

For now, the United States still retains an edge in quantum computing. Yet over the past decade, China has invested at least \$10 billion in quantum technology, roughly ten times as much as the U.S. government. China is working to build quantum computers so powerful that they will easily crack today’s encryption. The country is also investing heavily in quantum networks—a way of transmitting information in the form of quantum bits—presumably in the hope that such networks would be impervious to monitoring by other intelligence agencies. Even more alarming, the Chinese government may already be storing stolen and intercepted communications with an eye to decrypting them once it possesses the computing power to do so, a strategy known as “store now, decrypt later.” When quantum computers become fast enough, all communications encrypted through non-quantum methods will be at risk for interception, raising the stakes of achieving this breakthrough first.

China is also actively trying to catch up with the United States in synthetic biology. Scientists in this field are working on a range of new biological developments, including microbe-made cement that absorbs carbon dioxide, crops with an increased ability to sequester carbon, and plant-based meat substitutes. Such technology holds enormous promise to fight climate change and create jobs, but since 2019, Chinese private investment in synthetic biology has outpaced U.S. investment.

When it comes to semiconductors, China has ambitious plans, too. The Chinese government is funding unprecedented efforts to become a leader in semiconductor manufacturing by 2030. Chinese companies are currently creating what are known in the industry as “seven nanometer” chips, but Beijing has set its sights further, announcing plans to domestically produce the new generation of “five nanometer” chips. For now, the United States continues to outperform China in semiconductor design, as do U.S.-aligned Taiwan and South Korea. In October 2022, the Biden administration took the important step of blocking leading U.S. companies producing AI computer chips from selling to China as part of a package of restrictions released by the Department of Commerce. Yet Chinese companies control 85 percent of the processing of the rare-earth minerals that go into these chips and other critical electronics, offering an important point of leverage over their competitors.

A BATTLE OF SYSTEMS

The competition between the United States and China is as much a competition between systems as it is between states. In the Chinese model of civil-military fusion, the government promotes domestic competition and funds emerging winners as “national champions.” These companies play a dual role, maximizing commercial success and advancing Chinese national security interests. The American model, on the other hand, relies on a more disparate set of private actors. The federal government provides funding to basic science but largely leaves innovation and commercialization to the market.

For a long time, the trifecta of government, industry, and academia was the primary source of American innovation. This collaboration drove many technological breakthroughs, from the moon landing to the Internet. But with the end of the Cold War, the U.S. government grew averse to allocating funding for applied research, and it even lowered the amount devoted to fundamental research. Although private spending has taken off, public investments have plateaued over the past half century. In 2015, the share of government funding for basic research dropped below 50 percent for the first time since the end of World War II, having hovered around 70 percent in the 1960s. Meanwhile, the geometry of innovation—the respective role of public and private players in driving technological progress—has changed since the Cold War, in ways that have not always yielded what the country needs. The rise of venture capital helped accelerate adoption and commercialization, but it did little to address higher-order scientific problems.

The reasons for Washington's reluctance to fund the science that serves as the foundation of innovation power are structural. Innovation requires risk and, at times, failure—something politicians are loath to accept. Innovation can demand long-term investments, but the U.S. government operates on a single-year budget cycle and, at most, a two-year political cycle. Despite these obstacles, Silicon Valley (along with other hot spots in the United States) has still managed to encourage innovation. The American success story relies on a potent mix of inspiring ambition, startup-friendly legal and tax regimes, and a culture of openness that allows entrepreneurs and researchers to iterate and improve on new ideas.

That may not be enough, however. Government support has long played a critical role in jump-starting innovation in the United States, and research in technologies that seem outlandish now may prove critical in the not-too-distant future. In 2013, for example, the Defense Advanced Research Projects Agency invested in messenger RNA vaccines, working with the biotech company Moderna, which would later develop and deliver a COVID-19 vaccine in record time. But such examples are rarer than they should be.

Competition with China demands a reenergizing of the interplay among the government, the private sector, and academia. Just as the Cold War led to the creation of the National Security Council, today's tech-fueled competition should spur a rethinking of existing policymaking structures. As the National Security Commission on Artificial Intelligence (which I chaired) recommended, a new "technology competitiveness council," inspired by the NSC, could help coordinate action among private actors and develop a national plan to advance crucial emerging technologies. In a promising sign, Congress appears to have recognized the need for decisive support. In 2022, in a bipartisan vote, it passed the CHIPS and Science Act, which directs \$200 billion in funding for scientific R & D over the next ten years.

INVESTING IN THE FUTURE

As part of its effort to ensure that it remains an innovation superpower, the United States will need to invest billions of dollars in key areas of technological competition. In semiconductors, perhaps the most vital technology today, the U.S. government should redouble its efforts to onshore and "friend shore" supply chains, relocating them to the United States or friendly countries. In renewable energy, it should fund R & D for microelectronics, stockpile the rare-earth minerals (such as lithium and cobalt) needed for batteries and electric vehicles, and invest in new technologies that can replace lithium-ion batteries and offset China's resource dominance. Meanwhile, the rollout of 5G in the United States has been slow, in part because government agencies—most notably, the Department of Defense—control most of the high-frequency radio spectrum that 5G uses. To catch up with China, the Pentagon should open up more of the spectrum to private actors.

The United States will need to invest in all parts of the innovation cycle, funding not just basic research but also commercialization. Meaningful innovation requires both invention and implementation, the ability to execute and commercialize new inventions at scale. This is often the main stumbling block. Research in electric cars, for example, helped General Motors bring its first model onto the market in 1996, but it took two more decades before Tesla mass-produced a commercially viable model. Every new technology, from AI to quantum computing to synthetic biology, must be pursued with the clear goal of commercialization.

In addition to directly investing in the technologies that fuel innovation power, the United States must invest in the input that lies at the core of innovation: talent. The United States boasts the world's top startups, incumbent companies, and universities, all of which attract the best and the brightest from around the world. Yet too many talented people are prevented from coming to the United States by its outdated immigration system. Instead of creating an easy path to a green card for foreigners who earn STEM degrees from American schools, the current system makes it needlessly difficult for top graduates to contribute to the U.S. economy.

The United States has an asymmetric advantage when it comes to employing highly skilled immigrants, and its enviable living standards and abundant opportunities explain why the country has attracted most of the world's brightest AI minds. More than half of all AI researchers working in the United States hail from abroad, and the demand for AI talent still far exceeds supply. If the United States closes its doors to talented immigrants, it risks losing its innovative edge. Just as the Manhattan Project was led in large part by refugees and émigrés from Europe, the next American technological breakthrough will almost certainly rely on immigrants.

THE BEST DEFENSE

As part of its efforts to translate innovation into hard power, the United States must fundamentally rethink some of its defense policies. During the Cold War, the country designed various “offset” strategies to counterbalance Soviet numerical superiority through military strategy and technological innovations. Today, Washington needs what the Special Competitive Studies Project has called an “Offset-X” strategy, a competitive approach through which the United States can maintain technological and military superiority.

Given how much modern **militaries and economies** rely on digital infrastructure, any future **great-power war** is likely to start with a cyberstrike. The United States’ cyberdefenses, therefore, need a response time faster than humans’ reaction time. Having faced **constant cyberattacks** even in peacetime, the United States should armor itself with redundancy, creating backup systems and alternative paths for data flows.

What starts in cyberspace could **easily escalate into the physical realm**, and there, too, the United States will need to meet new challenges. To counter possible **swarm drone attacks**, it must invest in defensive artillery and missile systems. To improve battlefield awareness, the U.S. military should focus on **deploying** a network of inexpensive sensors powered by **AI** to monitor contested areas, an approach that is often more effective than a single, exquisitely crafted system. As human intelligence becomes harder to obtain, the United States must increasingly rely on the largest constellation of sensors of any country, ranging from undersea to outer space. It will also need to focus more on open-source intelligence, given that most of the world’s data today is publicly available. Without this **capability, the United States risks being surprised by its intelligence failures.**

When it comes to actual fighting, military units should be networked and decentralized to better outmaneuver opponents. Facing adversaries with rigid military hierarchies, the United States could gain an advantage by using smaller, more connected units whose members are adept at network-based decision-making, employing the tools of artificial intelligence to their advantage. For example, a single unit could bring together capabilities in intelligence collection, long-range missile attacks, and electronic warfare. The Pentagon needs to provide battlefield commanders with all the best information and allow them to make the best choices on the ground.

The U.S. military must also learn to integrate new technologies into its procurement process, battle plans, and warfighting. In the four years that I chaired the Defense Innovation Board, I was astounded by how difficult this was to do. A major bottleneck is the Pentagon’s burdensome procurement process: major weapons systems take more than ten years to design, develop, and deploy. The Department of Defense should look for inspiration in the way the tech industry designs products. It should build missiles the way companies now build electric cars, using a design studio to develop and simulate software, looking for innovations ten times as fast and as cost-effective as current processes. The current procurement system is especially ill suited for a future in which software primacy proves decisive on the battlefield.

The United States spends four times as much as any other country to procure military systems, but price is a poor metric for judging innovation power. In April 2022, Ukrainian forces fired two Neptune missiles at the Moskva, a 600-foot Russian warship, sinking the vessel. The ship cost \$750 million; the missiles, \$500,000 apiece. Likewise, China’s state-of-the-art hypersonic antiship missile, the YJ-21, could someday sink a \$10 billion U.S. aircraft carrier. The U.S. government should think twice before committing another \$10 billion and ten years to such a vessel. It often makes more sense to buy many low-cost items instead of investing in a few high-ticket prestige projects.

PLAYING TO WIN

In the **contest of the century**—the U.S. rivalry with China—the **deciding factor** will be innovation power. **Technological advances in the next five to ten years will determine which country gains the upper hand** in this world-shaping competition. The challenge for the United States, however, is that government officials are incentivized to avoid risk and focus on the short term, leaving the country to chronically underinvest in the technologies of the future.

If necessity is the mother of invention, **war is the midwife of innovation**. Speaking to Ukrainians on a visit to Kyiv in the fall of 2022, I heard from many that the first months of the war were the most productive of their lives. The United States’ last truly global war—World War II—led to the widespread adoption of penicillin, a revolution in nuclear technology, and a breakthrough in computer science. Now, **the United States must innovate in peacetime, faster than ever before. By failing to do so, it is eroding its ability to deter—and, if necessary, to fight and win—the next war.**

The alternative could be disastrous. Hypersonic missiles could leave the United States defenseless, and cyberattacks could cripple [collapse] the country's electric grid. Perhaps even more important, the warfare of the future will target **individuals** in completely new ways: authoritarian states such as China and Russia may be able to collect individual data on Americans' shopping habits, location, and even DNA profiles, allowing for **tailor-made disinformation campaigns** and even **targeted** biological attacks and **assassinations**. To avert these horrors, the United States needs to make sure it remains **ahead** of its technological competitors.

The principles that have defined life in the United States—freedom, capitalism, individual effort—were the right ones for the past and remain so for the future. These basic values lie at the foundation of an innovation ecosystem that is still the envy of the world. They have enabled breakthroughs that have transformed everyday life around the world. The United States started the innovation race in pole position, but it cannot rest assured it will remain there. Silicon Valley's old mantra holds true not just in industry but also in geopolitics: **innovate or die**.

1NC---Multilat CP

The United States federal government should

- engage in good-faith negotiations on an amendment to the Agreement on Trade-Related Intellectual Property granting the World Trade Organization the authority to strengthen and weaken dispute settlement protection of international intellectual property rights within the United States, including a requirement to **significantly strengthen patent protection by adopting a holistic and factually determined patent-eligibility test.**
- require compliance with the negotiated outcome
- prohibit significantly strengthening its protection of domestic intellectual property rights on the grounds that it violates maximum standards under the Agreement on Trade-Related Intellectual Property

CP solves the case and trade multilateralism

Peter K. Yu 11, Kern Family Chair in Intellectual Property Law & Director, Intellectual Property Law Center, Drake University Law School; Wenlan Scholar Chair Professor, Zhongnan University of Economics and Law; Visiting Professor, Centre for International Intellectual Property Studies, University of Strasbourg. Six Secret (and Now Open) Fears of ACTA, 64 SMU L. Rev. 975 (2011). July.
<https://scholarship.law.tamu.edu/cgi/viewcontent.cgi?article=1437&context=facscholar>

In addition, by **developing intellectual property protection outside** the traditional fora, like WIPO and **the WTO**, the negotiation of ACTA **undermined the stability of the international trading system and the preference for the multilateral process.** 612 It also alienated the trading partners of the ACTA negotiating parties, making it more difficult to undertake future multilateral discussions.

To some extent, it reminds us of a major shortcoming of **the unilateral approach embodied** in the Section 301 process. As Helen Milner wrote two decades ago of Special 301 and Super 301: Aggressive, bilateral reciprocity **violates central tenets of the postwar international trading system**. GATT upholds the principles of **multilateralism, nondiscrimination, and neutral dispute settlement** **Violations of international law by leading powers will induce other states to violate those laws as respect for them declines. Disregarding GATT norms will bring the entire system into question and may lead to its breakdown,** as U.S. actions did to the Bretton Woods monetary regime in the early 1970s. Since GATT has helped provide a stable, prosperous trading environment for forty years, ending it should not be done

lightly. Moving from a system of multilateral negotiation and dispute settlement to a bilateral one will increase the costs of negotiating trade liberalization and will greatly politicize the process. Undermining the GATT system in exchange for marginal improvements in the U.S. trade balance does not seem to be a rational strategy. 613 Likewise, Assafa Endeshaw expressed concern within the TRIPS context: The United States approach will work towards overthrowing any measure of success that the United States has achieved in placing intellectual property on an arguably 'international' pedestal (the TRIPs) after passing through long periods of bilateral arrangements. Consequently, the quiet overhaul that the international IP system has been subjected to through the TRIPS may now be in danger of collapse by the American insistence that it will interpret IP treaties and take any measures it deems appropriate, unilaterally and from its own national perspective. Each move of the United States to take IP matters throughout the world in its own hands will increasingly reduce the global significance of the TRIPS formula to a national system that has been outdated for quite some time. 614

Although both Professors Milner and Assafa made their observations on the ill-advised Section 301 sanctions, which have been found to be inconsistent with the WTO Agreement if taken before all remedies permissible under the WTO rules have been exhausted, these observations are highly relevant to the discussion of ACTA today and remind us of the high costs of the "pick and choose" approach. It took the European Union, Japan, and the United States a tremendous amount of time, effort, and energy to create the WTO and the TRIPS Agreement. 615 Such creation entailed a considerable amount of coordination, negotiation, and compromise between and among developed and less-developed countries. 616 Although the WTO remains far from satisfactory, and the TRIPS Agreement has yet to provide up-to-date protection to meet the needs of many developed countries and their intellectual property rights holders, the costs of undermining such a well-built system is not to be ignored.

To be certain, the European Union and the United States have complained about the unwillingness of less-developed countries to play ball. 617 As stated in Part I.B, the former has tried hard to advance discussion of enforcement issues at both the TRIPS Council and the ACE, at least in the image of its own laws. 618 The frustrations of these countries and their government officials are understandable. However, if countries always negotiate outside the forum when others refuse to play ball, the international trading system is unlikely to be developed the way it is today. In fact, the system owes much of its success to those great statesmen who work hard to create a diplomatic dialogue and to strike compromises among the parties. Diplomacy and statesmanship, after all, are as important as, if not more important than, bargaining.

It is also ironic that developed countries are now complaining about WIPO's limited ability to set international intellectual property enforcement norms when they are the ones who pushed such norm-setting discussions away from this particular organization to the GATT/WTO in the mid-1980s. 619 To those who are familiar with the TRIPS negotiating history, the developed countries' criticism of WIPO for its failure to take on a bigger role in enforcement seems rather disingenuous.

Moreover, by creating a new forum for intellectual property enforcement, ACTA would take away the opportunity to strengthen enforcement norms in the current regimes—whether under the WTO or WIPO. In fact, the creation of multiple fora with overlapping jurisdictions has created serious challenges for national governments. As Kimberlee Weatherall reminded us: There is a risk of confusion and fragmentation in this process, particularly, one would think, for government departments and enforcement bodies subject to multiple overlapping requirements found in multiple overlapping agreements. In a context where we want government to be more efficient, subjecting them to multiple sources of regulation is not likely to lead to happy results. What would we rather government be doing—actually encouraging innovation, or box ticking on their customs processes to check compliance with the multiple different obligations in different treaties? What should money be spent on—grants for artists or yet forms and bodies and meetings about counterfeiting? 620 It is important to remember, given the limited resources, not every country will have the ability to undertake discussions in multiple fora. The more fora there are, the more difficult it will be for less-developed countries to dedicate their efforts to strengthening enforcement norms in a particular forum.

Trade multilat is on the brink. Collapse causes extinction

Shiro **Armstrong 23**, Visiting Fellow, Research Institute of Economy, Trade and Industry. November 13. Professor, Crawford School of Public Policy, Australian National University Director, Australia-Japan Research Centre Director, East Asian Bureau of Economic Research "Countering Great-Power Trade Disruption" <https://www.rieti.go.jp/en/papers/contribution/shiro-armstrong/13.html> //

GREAT-POWER RIVALRY is undermining economic interdependence as a source of prosperity and security in East Asia. Collective international action to support the multilateral system could avoid turning economic interactions into a zero-sum game and a source of rivalry and conflict.

Economic interdependence, built on confidence in an open, rules-based multilateral system, has brought countries in East Asia closer together economically, socially and politically. Political rivalries and unresolved historical issues have been kept in check in Northeast and Southeast Asia by deepening economic integration founded on an embrace of open economic policies. The same cannot be said, however, for neighboring South Asia, as well as for many other regions.

Weaponized interdependence and the securitization of trade threatens to unwind the pacifying effects of economic exchange. The retreat of US leadership in the multilateral trading system from enforcer to spoiler, combined with economic coercion from China and US-China strategic competition, threaten the regional and global order. Uncertainty from the war in Ukraine, new technologies, outdated multilateral rules, climate change and recovery from the pandemic, together with the mismanagement of domestic social policies, have all contributed to US-China strategic competition. Many countries, especially in East Asia, including US allies, are faced with narrowed policy options and false choices between security and prosperity or between China and the US.

REJECTING DECOUPLING

The trajectory of strategic competition between the US and China is not inevitable. The interventions from Japan and Europe at the G7 summit in May demonstrate agency by the rest of the world. The G7 rejected an anti-coercion instrument aimed at China and reportedly favored by the US, which would have been a retreat from the rule of law globally and a step toward vigilante-mob justice. Instead, the G7 emphasized the importance of the multilateral system with the World Trade Organization (WTO) at its core, for economic resilience and economic security. Europe shifted Washington from decoupling to de-risking, potentially a significant de-escalation in the tussle between Washington and Beijing.

The protectionist stance and perceived national security interests of the US that drove policies to decouple advanced technology from China, with a “small yard and high fence” approach, presage a growing yard and a growing fence. The US’s extraterritorial, unilateral sanctions to achieve that decoupling have brought huge costs and increased risks for the rest of the world.

Discarding existing regional and global institutions because of their weaknesses and pursuing values-based co-operation may be politically expedient, but the approach fails to recognize the reality of a deeply interdependent global economy, especially in East Asia, and the necessity of interest-based global co-operation to overcome the biggest challenges faced by the global community, such as avoiding catastrophic climate change.

Decoupling is not realistic for East Asia. It has the most to lose from such an approach. Almost all supply chains go through China and decoupling would cause economic and political havoc. Efforts to decouple fail to recognize that a Chinese economy and society that is much less integrated into the global economy is one with far fewer constraints on it politically and is therefore much more of a security risk.

Preserving mutually beneficial economic exchange, or positive-sum engagement, is not just important for prosperity but for political security. Strategic competition that frames economic exchange as zero sum, that sees one country’s gain as a loss by another, has let loose geo-economic policies that involve pointing economic weapons inward, resulting in self-harm.

LEARN FROM HISTORY

The main elements of the current global economic order were forged during the throes of global warfare. The formulation of the Bretton Woods institutions was deeply influenced by the entanglement of economics and security in the 1920s and 1930s, a time when zero-sum economic competition in the Great Depression helped to fuel the national rivalries that propelled states into military conflict. The principles of the post-war order of multilateralism and liberalism were designed to avoid the

fundamental causes of insecurity and **conflict**. They were not the creation of economic technocrats naive to the very real security challenges.

The multilateral trading system, built on the principle of equal treatment — the most-favored nation (MFN) principle — **has kept the global economy open and acted to protect against economic shocks and aggression**. The large global economy with alternative buyers and suppliers has protected countries against trade stoppages. The primary lesson of Australia's 2020 experience of Chinese export bans for political purposes was that the **economic weapons China fired were blunted by the multilateral trading system**.

Deepening economic engagement and **strengthening rules**, rather than avoiding engagement, can **mitigate** and diffuse trade **risks**. Economic engagement builds national wealth and power — and when combined with multilateral rules, **broadens** the range of strategic policy **options** available to national policymakers.

Russia's strategic use of gas supplies against Europe is sometimes cited as a counterpoint to the argument that interdependence promotes political security. But Russia was not integrated into European supply chains, and European energy dependence on Russia is qualitatively different from economic interdependence in East Asia. **Interdependence underpinned by multilateralism effectively diffuses risks**.

THE LESSONS OF ASEAN

Nowhere is the power of multilateralism understood better, and exercised more instinctively, than in Southeast Asia's 10-member Association of Southeast Asian Nations (ASEAN). As the world transitions to a multipolar order, the principles of interstate interactions embedded in ASEAN's Treaty of Amity and Cooperation (TAC) can be strengthened and expanded to keep Southeast Asia open and coherent. The dialogue partners that have signed onto TAC include Australia, China, Japan, India and the US. Multilateralization of TAC among the dialogue partners is a long-term objective.

ASEAN has responded to its external circumstances with initiatives that include the Regional Comprehensive Economic Partnership (RCEP) and the ASEAN Outlook on the Indo Pacific (AOIP). Alongside the convening power of ASEAN, they demonstrate the utility of the grouping.

RCEP was brought into being at a time when protectionism was on the rise in Europe and the US. It importantly includes an economic co-operation agenda that has the potential to upgrade rules and standards over time and address new issues, including beyond narrowly economic issues, given the agreement has a ministerial and leaders' level process. There is an opportunity to keep the region free and open by working with ASEAN to build economic security through intensifying economic co-operation in RCEP, including with China.

Consistent with ASEAN's focus on inclusive arrangements that preserve centrality and an open approach to regionalism, a comprehensive regional security arrangement that affirms a commitment to multilateral economic rules and signs on to TAC principles will help secure a free, open, inclusive, prosperous and politically stable region. This frames a vision for the region in which ASEAN dialogue partners would shape their future in a way that references principles of crucial importance to prosperity and security.

Small- and medium-sized powers are protected by international rules and markets. Open, contestable markets constrain big powers that skirt established rules and use economic leverage, without thought for its ramifications. Even if big powers deviate from the rules, defending and extending the multilateral system is still the top priority for the rest of the world — not following larger powers down a self-destructive road.

THE WTO YET LIVES

Preservation of the multilateral system should be the priority for the global community to protect its economic and security interests. The WTO is **showing signs of life after** a package of agreements concluded in Geneva in 2022, including **the WTO's first multilateral agreement since** the Uruguay Round of the General Agreement on Tariffs and Trade (GATT) concluded in **1994**.

The strong emphasis on the multilateral trading system at the G7 summit in Hiroshima was preceded by Japan's entry into the Multi-Party Interim Appeal Arbitration Arrangement (MPIA) in March that **means the WTO's rules are enforceable between Japan and 52 other WTO members**. **WTO rules have not been enforceable since 2019 because of the US veto of the appointment of judges to the Appellate Body in the organization's dispute settlement system**.

The European Union led the creation of MPIA in 2020 with Australia, Canada, Singapore and, importantly, China, among the original members. **Trade ministers agreed in Geneva to make progress on restoring the dispute settlement system by the end of 2024, but more countries will need to join MPIA as insurance against US blackmail of the system during an election year**. Singapore is the only ASEAN member of MPIA.

The **WTO** is in need of reform. The rulemaking process and the trade **rules need to be updated**. It's no surprise that the areas of contention between economies — in the digital economy, industrial subsidies, investment and technology — are largely where there are no established rules.

Much of the rulemaking is happening outside of the WTO, in bilateral, regional and plurilateral agreements. These overlapping agreements are keeping the global economy from fracturing but rely on the WTO underpinning them, given the gaps in coverage, membership and rules. Regional agreements will have to be open and inclusive, otherwise they risk weakening the multilateral system, not strengthening it.

Digital rulemaking, for example, is happening in the WTO e-commerce plurilateral, the CPTPP, RCEP, DEA and DEPA. The bottom-up processes are guided by multilateral processes and principles in APEC and the Data Free Flow with Trust process that Japan initiated at the Osaka track alongside the G20 summit.

The Indo-Pacific Economic Framework (IPEF) excludes China by design, but the membership includes many countries that will want to avoid the exclusion of China. Despite all the problems with the WTO and the challenges it faces, neither the US nor China have walked away from it. In fact, the resolution of some trade disputes between Australia and China — with barley, for example — have been due to cases at the WTO, with both parties respecting the dispute settlement process. The global economy, including China, has a huge stake in the WTO's functioning.

The existing economic order is under threat but the trajectory of the global economy is not inevitable or predestined. The transition to a new multipolar order **does not require** another **world war**. Collective leadership can help shape Chinese and US behavior and avoid great power strategic competition from causing **catastrophic disorder**.

The biggest challenges that face the global community **require global co-operation** to overcome. The best hope of avoiding the **existential** risk of **climate change** will **require** China and the US to work together. It will also require **technological progress** and open **exchange** of goods, capital and technology. Technology decoupling between the world's two largest producers of knowledge and innovation will significantly raise the costs of decarbonizing the global economy. That overwhelming positive-sum interest may be the best hope of avoiding zero-sum games that lead to a prisoner's dilemma equilibrium of much lower living standards and extreme insecurity.

Economic interests and co-operation are usually the first casualty of geopolitical tensions and heightened security challenges. But to retreat to a world in which trade is frustrated from delivering its positive-sum outcomes risks unwinding interdependence and the prosperity and security that it provides.

The Judiciary

Scenario one

Alt Cause — judge shopping ruins court stability and credibility.

Kanu 24 — Hassan Ali Kanu, Staff Writer for *The American Prospect*, 2024 ("America's Fifth Circuit Problem," *The American Prospect*, April 15th, Available Online at <https://prospect.org/justice/2024-04-15-americas-fifth-circuit-problem/>, Accessed 09-08-2024)

Corporate America and conservative activists agree: We live in the United States of Texas, Louisiana & Mississippi.

These states have increasingly played host to major federal lawsuits implicating national policy, primarily in the Northern District of Texas, where certain divisions have just one or two judges, appointed by Republican presidents. The Amarillo division, for example, is only assigned one judge, and so cases there flow unilaterally to ultraconservative Donald Trump appointee and anti-abortion activist Matthew Kacsmayk. In the Fort Worth division, you get a choice of two judges: George W. Bush appointee Reed O'Connor, or Trump judge Mark Pittman.

All of these **judges are** reliable **champions for a radically conservative** vision of the **law**. When their rulings are appealed, they end up at the just-as-conservative Fifth Circuit, also stacked with partisan

ideologues. This has generated a pipeline that fast-tracks right-wing rulings to the Supreme Court—and creates endless delays for policies that protect and benefit the public.

The underlying strategy is known as judge shopping, and it's key to the right-wing project to politicize and capture the judiciary in order to forcefully enact unpopular policy priorities. According to the public interest group Accountable.US, since January 2017, 63 percent of all U.S. Chamber of Commerce lawsuits challenging federal regulations have been filed at district courts within the Fifth Circuit's boundaries.

The Fifth Circuit is apparently doubling down even as backlash mounts, essentially litigating against other federal courts and even some of its own circuit colleagues over the right to hear high-stakes, politically charged cases. It's an unprecedented development that highlights the broader question of what can be done about partisan, rogue judges operating in service to raw power.

ON APRIL 5, REPUBLICAN-APPOINTED FIFTH CIRCUIT JUDGES issued an extraordinary order to Judge Pittman in a lawsuit involving the Consumer Financial Protection Bureau's effort to cap late fees for credit card issuers. The U.S. Chamber of Commerce and various bank lobbying groups sued to block the cap from taking effect.

Despite his usual ideological approach, Pittman had transferred the Chamber's case to the federal district court in D.C. a week earlier, explaining that the dispute didn't have any substantial connections to the Fort Worth area. On the other hand, Pittman explained, the rule was developed in D.C., by agencies and employees stationed there, and most of the plaintiff organizations and their lawyers are also based in D.C.

That ruling implicitly acknowledged broad criticism of judge shopping within the Fifth Circuit.

And Pittman's ruling seemed technically unassailable: Such a transfer can only be overturned if it's an abuse of discretion and "patently erroneous"—where the lower court can't even offer a coherent explanation, in other words. And that basically never happens.

"Everybody has known forever that once a case is transferred out there's nothing you can do about it," said David Coale, an appellate lawyer at Lynn Pinker Hurst & Schwegmann in Dallas.

But that didn't matter to the Fifth Circuit. A majority of the Republican-appointed appellate judges asked Pittman last week to request a retransfer of the case from D.C. back to Texas.

The Fifth Circuit may have stretched even the bounds of its own lawful authority in order to do so, as the dissenting judge effectively conceded. The "new proposition of law created by the majority is incompatible with district court discretion over [its own] docket management and prudent policing of forum shopping," Judge Stephen Higginson wrote.

Moreover, Judge Don Willett, part of the appellate panel, holds significant amounts of stock in Citigroup, which could benefit from a ruling blocking the late fee cap. (He's not even the first judge involved in the case found to have credit card company stock holdings.) Willett asked for briefings from both sides of the case on whether he should recuse. Yet the Chamber argued that Willett should not recuse because the case "would have limited effect on the bottom line of issuers"—which makes you wonder why they're bothering to sue to begin with.

Unfortunately, the Fifth Circuit's aggressive moves to retain jurisdiction have already succeeded. Judge Amy Berman Jackson, who received the CFPB case from Judge Pittman, terminated the case in the D.C.

district court, effectively sending it back to Texas. Judge Jackson did say in her defense that “the Court is not inclined to ‘disregard’ a case on its docket, and it has considerable discretion to supervise its own cases.” But she then used that discretion to simply transfer a case that was rightfully in her court to another, much less appropriate venue, explaining only that the district court in Texas had docketed the case. (If it isn’t obvious, Jackson’s own court had also formally accepted and placed the case on its docket.)

In short, Jackson saved the Fifth Circuit the embarrassment of having a case wrested away—an extremely rare situation that played out in 2022, when a district judge in New Jersey refused to retransfer a case about the legality of instructions for 3D-printed guns back to the ultraconservative circuit.

Incredibly, the Fifth Circuit is embroiled in a number of other ongoing venue fights. A suit filed by Elon Musk’s SpaceX, challenging the constitutionality of the National Labor Relations Board, was transferred to a district court in California by district court judge Rolando Olvera, a rare Obama appointee in Texas. The Fifth Circuit hasn’t let go of the case, even after a three-judge panel initially denied SpaceX’s request to take it back to Texas. A different Texas judge dismissed a case involving the pharmaceutical industry’s challenge of the Inflation Reduction Act’s law allowing Medicare to negotiate drug prices, due to improper venue. But the Fifth Circuit is hearing an appeal to that dismissal, hanging on to the case, with oral arguments scheduled for the end of the month.

Historically, judge shopping has been universally condemned, for the obvious reasons that it erodes impartiality and denigrates the public’s trust in the courts. That consensus includes the Supreme Court; the Biden administration; Congress; the American Bar Association; and virtually every legal scholar that has studied the issue. Chief Justice John Roberts endorsed random assignment of cases in his 2021 annual report on the federal courts.

But now, the judges most implicated in judge shopping are openly embracing the practice, potentially even overstepping their authority in order to carry business groups’ water. Even worse, they have the full support of the Republican Party leadership.

LAST MONTH, REPUBLICAN LEADERS IN CONGRESS PUBLICLY URGED appellate judges around the country to resist adopting rules that would prevent judge shopping, contradicting recent guidelines issued by the Judicial Conference, which governs the federal judiciary. Those guidelines would strengthen procedures for randomly assigning cases of national importance to judges—largely an attempt to address the gamesmanship at the Fifth Circuit.

The move should have been entirely uncontroversial, but it met fierce resistance from Republicans like Sen. Mitch McConnell (R-KY), one of the prime architects of the right-wing takeover of the courts. McConnell and his allies have urged chief judges across the country not to change their case assignment practices. And some of the more strident conservative judges in the country joined in, making highly unusual public statements criticizing the Judicial Conference’s policy, and suggesting they would in fact do nothing to address judge shopping.

“The Republican Party has turned into a pro-judge shopping lobby,” said Paul Gugliuzza, a professor at Temple University Beasley School of Law who has studied the issue. “Until a few weeks ago, the idea that one party shouldn’t get to pick the individual judges that hear their cases was a principle you didn’t even have to defend.”

The cracks in the consensus around judge shopping have been enabled largely by the same Republican politicians who have led their party's movement to appoint ideologues and conservative partisans, rather than good-faith actors, to the federal courts. Of course, a number of those judges now sit on the U.S. Supreme Court, which shares much of the responsibility here.

Read Rebecca's cards on email

Democracies aren't peaceful — violence and aggression is guaranteed.

Chiba & Gartzke 21 — Dania Chiba, Associate Professor of Political Science in the Department of Government and Public Administration at the University of Macau, Ph.D in Political Science from Rice University, LL.M in Jurisprudence and International Relations from Hitotsubashi University, Erik Gartzke, Professor of Political Science at the University of California, San Diego, PhD in Political Science from the University of Iowa, 2021 ("Make Two Democracies and Call Me in the Morning: Endogenous Regime Type and the Democratic Peace," *Office of Naval Research*, February 19th, Available Online at <https://dainachiba.github.io/research/make2dem/Make2Dem.pdf>, Accessed 09-07-2024)

The democratic peace—the observation that democracies are less likely to fight each other than are other pairings of states—is one of the most widely acknowledged empirical regularities in international relations. Prominent scholars have even characterized the relationship as an empirical law (Levy 1988; Gleditsch 1992). The discovery of a special peace in liberal dyads stimulated enormous scholarly debate and led to, or reinforced, a number of policy initiatives by various governments and international organizations. Although a broad consensus has emerged among researchers regarding the empirical correlation between joint democracy and peace, disagreement remains as to its logical foundations. Numerous theories have been proposed to account for how democracy produces peace, if only dyadically (e.g., Russett 1993; Rummel 1996; Doyle 1997; Schultz 2001).

At the same time, peace appears likely to foster or maintain democracy (Thompson 1996; James, Solberg, and Wolfson 1999). A vast swath of research in political science and economics proposes explanations for the origins of liberal government involving variables such as economic development (Lipset 1959; Burkhart and Lewis-Beck 1994; Przeworski et al. 2000; Acemoglu and Robinson 2006; Epstein et al. 2006) and inequality (Boix 2003), political interests (Downs 1957; Bueno de Mesquita et al. 2003), power hierarchies (Moore 1966; Lake 2009), third party inducements (Pevehouse 2005) or impositions (Peceny 1995; Meernik 1996), geography (Gleditsch 2002b), and natural resource endowments (Ross 2001), to list just a few examples. Each of these putative causes of democracy is also associated with various explanations for international conflict. Indeed, some as yet poorly defined set of canonical factors may contribute both to democracy and to peace, making it look as if the two variables are directly related, even if possibly they are not.

We seek to contribute to this literature, not by proposing yet another theory to explain how democracy vanquishes war, but by estimating the causal effect of joint democracy on the probability of militarized disputes using a quasi-experimental research design. We begin by noting that some of the common causes of democracy and peace may be unobservable, generating an endogenous relationship between the two. Theories of democracy and explanations for peace are at a formative state; it is not possible to utilize detailed, validated and widely accepted models of each of these processes to assess their

interaction. Indeed, to a remarkable degree democracy and peace each remain poorly understood and weakly accounted for empirically, despite their central roles in international politics.

We address the risk of spurious correlation by applying an instrumental variables approach. Having taken into account possible endogeneity between democracy and peace, we find that joint democracy does not have an independent pacifying effect on interstate conflict. Instead, our findings show that democratic countries are more likely to attack other democracies than are non-democracies. Our results call into question the large body of theory that has been proposed to account for the apparent pacifism of democratic dyads.

Scenario two

Patents Deck Innovation — empirics, downstream blocking, and patent trolls. First-mover advantage solves innovation independently.

Boldrin & Levin 12 — Michele Boldrin, M.A. of Science in Economics from the University of Rochester, NY, former Marc Rich Professor of Economics at Universidad Carlos III de Madrid, former economic advisor to the Bank of Japan, David K. Levine, Robert Schuman Center for Advanced Study Joint Chair at the European University Institute, John H. Biggs Distinguished Professor of Economics Emeritus at Washington University in St. Louis, 2012 (“The Case Against Patents,” *The Federal Reserve Bank of St. Louis*, September, Available Online at <https://s3.amazonaws.com/real.stlouisfed.org/wp/2012/2012-035.pdf>, Accessed 08-19-2024)

[edited for ableist language]

1. Introduction

The case against patents can be summarized briefly: there is no empirical evidence that they serve to increase innovation and productivity, unless the latter is identified with the number of patents awarded – which, as evidence shows, has no correlation with measured productivity. This is at the root of the “patent puzzle”: in spite of the enormous increase in the number of patents and in the strength of their legal protection we have neither seen a dramatic acceleration in the rate of technological progress nor a major increase in the levels of R&D expenditure – in addition to the discussion in this paper, see Lerner [2009] and literature therein. As we shall see, there is strong evidence, instead, that patents have many negative consequences. Both of these observations, the evidence in support of which has grown steadily over time, are consistent with theories of innovation that emphasize competition and first-mover advantage as the main drivers of innovation and directly contradict “Schumpeterian” theories postulating that government granted monopolies are crucial in order to provide incentives for innovation. The differing predictive and explanatory powers of the two alternative classes of models persist when attention is shifted to the historical evidence on the life-cycle of industries. The initial eruption of small and large innovations leading to the creation of a new industry – from chemicals to cars, from radio and TV to personal computers and investment banking – is seldom, if ever, born out of patent protection and is, instead, the fruits of highly competitive-cooperative environments. It is only after the initial stages of explosive innovation and rampant growth end that mature industries turn toward the legal protection of patents, usually because their internal growth potential diminishes and the industry structure become concentrated.

A closer look at the historical and international evidence suggests that while weak patent systems may mildly increase innovation with limited side-effects, strong patent systems retard [diminish] innovation with many negative side-effects. Both theoretically and empirically, the political economy of government

operated patent systems indicates that weak legislation will generally evolve into a strong protection and that the political demand for stronger patent protection comes from old and stagnant industries and firms, not from new and innovative ones. Hence the best solution is to abolish patents entirely through strong constitutional measures and to find other legislative instruments, less open to lobbying and rent-seeking, to foster innovation whenever there is clear evidence that laissez-faire under-supplies it.

2. Theory and Practice of Patents and Innovation

There is little doubt that providing a monopoly as a reward for innovation increases the incentive to innovate. There is equally little doubt that granting a monopoly for any reason has the many ill consequences we associate with monopoly power – the most important and overlooked of which is the strong incentive of a government granted monopolist to engage in further political rent-seeking to preserve and expand his monopoly or, for those who do not yet have it, to try obtaining one. Said differently: while the positive impact of patents is the straightforward partial equilibrium effect of increasing to the monopolistic level the profits of the successful innovator, the negative one is the subtler general equilibrium effect of reducing everybody else's ability to compete while increasing for everyone the incentive to wastefully lobby.

In the long run, though, even the positive partial equilibrium effect may be more apparent than real: the existence of a large number of monopolies due to past patent grants reduces the incentives for innovation as current innovators are subject to constant legal action and licensing demands from earlier patent holders. The recent – and largely successful – efforts of Microsoft to impose a licensing fee on the large and expanding Android phone market is but one case in point. With the exception of Motorola Mobility all the handset manufacturers have agreed to the fee, and Motorola has recently lost the first battle – fought not in court but in the more receptive domain of the U.S. International Trade Commission.¹

The basic problem with the patent system – the downstream-blocking effect of existing monopoly grants on future innovation – is greatly increased because modern products are made up of so many different components. The Microsoft-Motorola example is a good illustration as a licensing fee on Android mobile phones is being charged by Microsoft solely over a patent involving the scheduling of meetings – a trivial and rarely used feature of modern smart-phones. This is but one of many thousands of patented “ideas” used in a modern smart-phone, and each owner of each patent potentially can charge a licensing fee. Hence, the main dynamic general equilibrium effect of a patent system is to subject future inventions to a gigantic hold-up problem: with many licenses to be purchased and uncertainty about the ultimate value of the new innovation each patent holder, in raising the price of his “component”, imposes an externality on other patent holders and so charges a higher than efficient licensing fee. Boldrin and Levine [2005] and Llanes and Trento [2009] have explored the theory, many case-studies involving patents – and other fractionated ownership problems – can be found in Heller [2008].

A second widely cited benefit of patent systems – although not so much in the economics literature – is the notion that patents are a substitute for socially costly trade secrecy and improve communication about ideas. From a theoretical point of view the notion that patents are a substitute for trade secrecy fails even in the simplest model. If a secret can be kept for N years and a patent lasts M years then an innovator will patent exactly when $N \leq M$. Hence, only those things will be patented for which the secret would have emerged before the patent expired, while those for which the secret can be kept will not be patented. The same remains true in more complex and realistic models such as that of Boldrin and Levine [2004], see also Ponce [2007]. It is also the case that modern “disclosure” in patents is negligible – it is essentially impossible to build a functioning device or software program from a modern patent application, a fact which is especially clear since some patented ideas do not – and cannot – work. A case in point is the patent for moving information through the fifth dimension.²

A more subtle point is that secrecy may bias the type of inventive activity away from innovations that are not easily kept secret to those that can be, viceversa for patents. There is historical evidence from Moser [2004, 2005] in this direction. In 19th Century expositions of inventions, while countries without patent systems had overall rates of innovation similar to those with a patent system, they did specialize in innovations which were more easily kept secret. How strong this bias would be if no countries had patent systems – that is, whether it is true that patents encourage innovations that would not otherwise be made, or if they just shift the location of innovation from country to country – is not known.

The related idea that patents somehow improve communication about ideas – a notion key to the “public-private” partnership between governments and private research organization in which the government funds the research and then gives the private organization a monopoly over anything developed in the course of research – is backed neither by theory or evidence. It is impossible to study the history of innovation without recognizing that inventors and innovators exchange ideas as a matter of course and that secrecy occurs, in those cases in which it occurs, only in the final stages of an innovation process, when some ambitious inventors hope to corner the market for a functioning device by patenting it. A good case in point is that of the Wright brothers, who made a modest improvement in existing flight technology which they kept secret until they could lock it down on patents, then used their patents both to monopolize the U.S. market and to prevent innovation for nearly 20 years. This is discussed in Shulman [2003]. The role that Marconi and his patent played in the development of the radio is altogether similar – see Hong [2001] – as are innumerable others. At the opposite extreme we have, again among many, the example of the Cornish steam engine discussed in Nuvolari [2004a, b]. Here engineers exchanged nonpatented ideas for decades in a collaborative effort to improve efficiency. The modern and highly successful open source software movement is a more contemporary example of how collaboration and exchange of ideas thrives absent intellectual property. How much public benefit of the various patented – and never-the-less secret – pieces of the Microsoft operating system has occurred?

On the other side of the coin, the rationale for patent systems is weak. In most industries the first mover advantage and the competitive rents it induces are substantial without patents. Again: the smartphone industry – laden as it is with patent litigation – is a case in point.³ Apple derived enormous profits in this market before it faced any substantial competition. The first iPhone was released on June 29, 2007. The first serious competitor, the HTC Dream (using the Android operating system) was released only on October 22, 2008. By that time over 5 million iPhones had been sold, and sales soared to over 25 million units during the subsequent year, while total sales of all Android based phones was less than 7 million. In the tablet market the iPad still has no serious competitor despite having been introduced on April 10, 2010. While it is hard to prove this delayed imitation would have occurred also in the complete absence of patents, it is a fact that Apple did not try to use patents to prevent the Android phones from coming into its market and the subsequent “patents’ fight” has been taking place largely after 2010, something that the Boldrin and Levine [2004] model predicts. More to the point, companies typically instruct their engineers developing products to avoid studying existing patents so as to be spared subsequent claims of willful infringement, which raises the possibility of having to pay triple damages. Based on sworn testimony by Google’s chief of Android development in Oracle vs. Google (see for example Nicolai [2012]) the engineers that developed Android were unaware of Apple (or other) patents, and so were unlikely to have been helped by them.⁴

How valuable financially, for Apple, was the delay in the Android phones entry? Based largely on the fact that Apple has kept its first mover advantage in spite of a large imitative entry in this market, the value of Apple stock – during a severe market downturn – has gone up by a factor of approximately five. While there may have been some delay in competition due to Apple’s threat – since executed – of patent litigation, the fact is that similar but less successful devices had been available for a number of years before Apple finally cracked the market. The market for software and hardware may be viewed as a somewhat special case. Generally the fixed cost of producing software is low – although it is estimated that Apple spent 150 million USD developing the iPhone. This, however, pales in comparison to the cost of developing new medicines – which is estimated to have a present value of closer to 1 billion USD – the same way it does in front of that for developing a new model of automobile, which is in the same range. Interestingly it is also true that – according to both survey and anecdotal evidence – patents play an important role in encouraging innovation in the pharmaceutical industry while playing a minor one in that of cars, insofar as new components and even plants are often developed by consortia or joint-ventures of otherwise fiercely competitive producers marketing different automobile’s brands. The relevance of patents in the pharmaceutical industry – then, and contrary to “Schumpeterian” theories – is most likely not due to the high fixed costs but rather the fact that disclosure in the case of drugs is more meaningful than in that of cars and most other products. The chemical formula and the efficacy of the cure as established by clinical trials are available to competitors essentially for free and it is the second (a public good, privately produced due to a political choice) that accounts for about 80% of the initial fixed cost. On the other side, the downstream cost of monopoly pricing of pharmaceutical products is much higher for life-saving drugs, and the cost of monopoly pricing of other

pharmaceutical products is also quite high. Hence various economists, holding differing views about intellectual property, have nevertheless argued that if government intervention is indeed needed in this market a system of prizes would be far superior to the existing system of monopolies.

To understand patents in practice – and in particular why the poorly named “Schumpeterian” theories are not relevant to a proper understanding of either their effect upon, or the determinants of, innovative activity – it is necessary to examine the lifecycle of industries (see for example Jovanovich and McDonald [1994] and Scherer [1990]). Typically a new, hence innovative, industry begins with a competitive burst of entries through which very many innovators try hard to get their products to market. In these early stages, there are many firms bringing different versions of the new product to the market – just think at the American auto industry in the early twentieth century or the software industry in the 1980s and 1990s – demand for the new product grows rapidly and the latter is also rapidly improved. The price elasticity of demand is consequently high: what is important is not to dominate the market but rather try to get your own products quickly to it and to reduce costs. What this means is that your cost-reducing innovation is good for me in the same way that my cost-reducing innovation is good for you, hence let us all imitate each other and compete in the market. As the industry matures, demand stabilizes and becomes much less elastic; the scope for cost-reducing innovations decreases, the benefits of monopoly power grow and the potential for additional product innovation also shrinks. Typically there is a “shake-out” in which many firms either leave the industry or are bought out. The automobile industry is a classical historical example but the much more recent “bursting” of the dotcom “bubble” is, in fact, one we may recall better and that makes this point even more forcefully. At this stage rent-seeking does become important and patents are widely used to inhibit innovation, prevent entry, and encourage exit. If we look at patent litigation in practice – as is predicted by theories such as that of Boldrin and Levine [2004] – it takes place when innovation is low. The dead hand of dying institutions – Texas Instruments was famous for this and now we have the example of Microsoft – gets hold of the industry as they attempt to tax consumers, new entrants and any potential competitor. Far from being encouraged, when an industry matures innovation is blocked by the ever increasing appeal to intellectual property protection on part of the insiders.

It is somewhat conventional to think of welfare losses from distortions as small – the well known arguments about the small cost of the business cycle and of the general idea that welfare triangles due to monopoly power are small, being paradigmatic cases in point. With the monopolies of patents the opposite is the case. The conventional view starts with competitive equilibrium and observes that welfare losses from small price variations are quadratic as a function of the price, hence grow very slowly as the price increases. In the case of full monopoly – as is the case with patents – we are not interested in small price deviations from competition but rather we are interested in pricing near the top of the profit function. Witness for example the fact that patented pharmaceutical products often sell for hundreds of times the marginal cost of production as some astonishing pricing differences between the US and the European markets show. Here social loss increases nearly linearly with prices, while the deviation of profits from the maximum are quadratic. Hence small price increases have a only a small effect on profits – although still worthwhile from the perspective of the monopolist – with substantial social loss. It is impossible to observe the behavior of modern IP monopolists without recalling this theoretical prediction. Most of the copyright wars revolve around measure to prevent piracy, empirically a relatively minor factor as far as profits of media corporations are concerned (see for example Sinha, Machado, Sellman [2010], Danaher, Dhanasobhon, Smith and Rahul, [2010], Sanchez [2012]). In the case of patents, and particularly pharmaceutical patents, the situation is even more severe. We have made mention of the loss of human life due to the pricing of AIDS drugs. More revealing is the empirical study of the Quinolones family of drugs (Chaudhuri, et al. [2006]). It measures the economic consequences to India of the introduction of pharmaceutical patents for this family of drugs and concludes that the consequence to third world India will be nearly 300 million USD in welfare losses – while the gain to the first world pharmaceutical companies will be less than 20 million USD. The cost of litigating patents is not insubstantial either. Bessen and Meurer [2009] used stock market event studies to estimate the cost of patent litigation: they estimate that during the 1990s it rose substantially until at the end of the period it constitutes nearly 14% of total R&D costs. A related but more difficult to quantify phenomenon is the rise of uncertainty caused by the legal system. A case in point are the NTP Inc. patents that were used to threaten the Blackberry network with a shutdown. In 2006, posed on the edge of a cliff, Research in Motion, the producer of Blackberry, agreed to pay \$ 612.5 million to license the patent in question from NTP. The patent was later invalidated by the court – but Blackberry of course did not get their money back. Further details about this and other NTP-induced cases are available on the Wikipedia’s page about NTP. Here the bad behavior of a single judge cost Blackberry more than half a billion dollars. In this setting it is no surprise that patent trolls hope to get rich quickly.

Arms Races and Patent Trolls

Patenting has exploded over the last decades. In 1983 in the U.S. 59,715 patents were issued against 105,704 applications; by 2003, 189,597 patents were issued against 355,418 applications and, even in a slow growth year like 2010, 244,341 new patents were approved. In less than thirty years, the flow of patents roughly quadrupled. By contrast, neither innovation nor R&D expenditure have exhibited any particular upwards trend, not to speak of factor productivity. While patent litigation has increased, **few patents are actively used**. Patent litigation typically involves dying firms, that have accumulated huge stockpile of patents but are no longer able to produce marketable products, suing new and innovative firms. A once proud firm – one of the first producers of microchips, and who in our generation can forget their first TI calculator – Texas Instruments was unable to make the transition to the PC revolution and became, for a while, the symbol of a dying company trying to stay alive by suing the newcomers. In more recent times, Microsoft has become the chief among the patent trolls. Once the giant of the software industry Microsoft has been unable to make the leap to portable devices such as telephones and tablet PCs. Unable to create and produce for the marketplace, Microsoft now attempts to claim a share of the profits Google generates in this market through patent litigation. A firm that when it was young and innovative had a strong position against software patents – Bill Gates said in 1991: “If people had understood how patents would be granted when most of today's ideas were invented and had taken out patents, the industry would be at a complete standstill today...A future start-up with no patents of its own will be forced to pay whatever price the giants choose to impose.” – now lobbies in Europe and Asia for the introduction of software patents, which it already obtained in its home country. To learn more about the actual effect of patents in the real world, let us consider the response of Google to being pursued legally by a large competitor. One response is their recent purchase of Motorola Mobility. Motorola, like TI and Microsoft, is a once proud company that has fallen on hard times. Why the interest of Google? It is buying Motorola Mobility for its patent portfolio. Not for the ideas and innovations in that portfolio – few if any changes or improvements to Google's Android operating system will result from the ownership or study of these software patents. The purpose of obtaining this patent portfolio is purely defensive: it can be used to countersue Apple and Microsoft and blunt their legal attack on Google. Talk to anyone in the software industry – or any industry except pharmaceuticals or medical instruments. Read the Levin et al. [1987] and the Cohen et al. [2000] surveys of R&D managers. **The vast bulk of patents are not only useless, they don't represent innovation at all. They are part of an arms race. Any successful large company needs a large portfolio of patents to fend off potential lawsuits by rivals and by patent trolls.**

One could argue that purely defensive patenting is pretty harmless – after all it costs only about \$40,000 to file a successful patent application, and doing it on a large scale may make it cheaper. However the acquisition of large patent portfolios by incumbents creates huge barriers to entry. We see this in the smart-phone market. Here Apple is the market leader and Microsoft is unable to produce a product that appeals to consumers. Each are incumbent firms with a large patent portfolio. The new entrant and innovator – and it should be pointed out that while Android phones are imitations of iPhones in many respects, their integration with the web is vastly superior to what is promised by Apple in its next generation of iPhones – is Google. Google is a relatively new firm and while wealthy does not have a large defensive patent portfolio. Hence we see both Apple and Microsoft attacking Google with patent litigation. **Apple in an effort to keep the market to itself for a few more years; Microsoft because it is better to get a share of Google's revenue than be shut out of the market completely. The actual economic value of Microsoft's patents can be measured by their inability to produce a product that occupies more than a small corner of a large and growing market**

Patents deck overall R+D investment.

Stallman 13 — Richard Stallman, Founder and President of the Free Software Foundation, holds a BA from Harvard University, Founder of the GNU operating system, keynote speaker at the Loyola, 2013 (“Patent Law Is, at Best, Not Worth Keeping,” Chicago Law Journal, vol. 45, no. 2, pg. 393-4, Winter, Available Online at <https://lawecommons.luc.edu/cgi/viewcontent.cgi?article=1420&context=luclj>, Accessed 08-21-2024)

Even the private investment in research is not necessarily increased by patents. Economic modeling by James Bessen shows that, taking account of the effects of incremental innovation, patents can discourage investment in research.²³ In the simplest case of this problem, the patent holder has no incentive to do research in improving the patented product because the patent holder already has a monopoly. Other companies have no incentive to do research in improving that product because they would not be allowed to sell the improved version.²⁴

The US is ahead in the tech race — talent and investment prove.

Sheng & Geng 24 — Andrew Sheng, distinguished fellow at the Asia Global Institute at the University of Hong Kong, and Xiao Geng, Chairman of the Hong Kong Institution for International Finance, professor and Director of the Institute of Policy and Practice at the Shenzhen Finance Institute, 2024 (“China Needs a Better Innovation Ecosystem,” Project Syndicate, April 2nd, Available Online at <https://www.project-syndicate.org/commentary/china-innovation-ecosystem-great-power-competition-with-us-by-andrew-sheng-and-xiao-geng-2024-04>, Accessed 08-18-2024)

The United States has long dominated much of the knowledge economy, not least owing to an innovation environment that has proved highly attractive to foreign talent. But China has been watching carefully, adapting crucial lessons from the US to the local context. And its efforts are bearing fruit.

America’s lead is significant. In 2000-19, the US had far more Fortune Global 500 companies than any other country. In 2000, the US boasted 179, compared to 107 for Japan and just ten for China. Though China surpassed the US in 2020, the US companies have retained an absolute advantage in market capitalization, profitability, and technological innovation.

American tech giants — such as Microsoft, Apple, and Nvidia — are especially dominant, having expanded faster internationally and outsourced their operations more broadly than their counterparts from China, the European Union, Japan, and India. Crucially, they have also attracted more of the world’s talent.

In fact, the US is the top destination for global talent, accounting for nearly half of the highly skilled immigrants entering OECD countries. According to the QS World University Rankings, one-third of the world’s top 100 universities, and four of the top ten, are in the US. These elite universities attract talented non-US students, who are likely to remain after graduation to start their careers.

As a result, the US is now home to one-third of the world’s high-skilled immigrant population. And this population is playing a leading role in shaping America’s business and innovation environment. For example, Indian CEOs oversee American companies — including leading firms like Adobe, Alphabet and its Google subsidiary, Microsoft, and IBM — that are collectively valued at \$6 trillion.

Chinese cities have been working hard to replicate this success. Shenzhen is a case in point. Within living memory, this southern Chinese city was a small fishing village, with few world-class universities nearby. But since 2000, the city has been transformed into a leading tech hub, thanks to its support for local universities and the establishment of satellite campuses by highly rated universities such as Peking University HSBC Business School, Tsinghua University, and the Chinese University of Hong Kong.

Cities attract top talent by offering an appealing lifestyle, a supportive business environment, and strong supply-chain connectivity. Industrial parks — which are being embraced by many developing economies — can bolster this effort, though establishing them is no easy feat, as they require talent in tech, corporate management, and government services, as well as supporting services in areas like finance, law, and logistics.

Financing is another crucial piece of the puzzle. A thriving innovation ecosystem cannot exist if high-tech firms cannot get the funding they need to grow. But for Chinese firms, it is becoming increasingly

difficult to secure foreign investment. Since 2018, hundreds of Chinese companies – including Huawei and BYD – have confronted new sanctions or restrictions from the US and Europe. Many multinationals have decided to “decouple” from Chinese supply chains in order to comply with home-country regulations, despite the higher production costs this implies.

Yet foreign capital is not essential. In Silicon Valley, innovative tech companies turn to private-equity (PE) and venture-capital (VC) funds to finance their growth. These US funds are constantly investing in the next generation of “unicorns” (private companies with a valuation above \$1 billion), often using their gains from the last generation. The National Venture Capital Association reports that about 80% of US PE funds invest in companies in the growth and development stages.

But in China (including Hong Kong), such funds tend to be smaller than their counterparts in the US: in 2021, the average PE fund size in the US approached \$1 billion, compared to some \$20 million in China. Moreover, Chinese PE/VC funds lack experience and specialist knowledge when it comes to frontier technologies.

This implies a serious challenge for China. From a macro-policy perspective, PE/VC funds are critical catalysts of supply-chain upgrading through new technologies and management knowhow. So, as Chinese companies seek to adopt new technologies, increase productivity, expand, and develop overseas markets, much of their risk capital will be financed primarily by these funds, rather than conventional banks or public financial markets.

But tech investments tend to have shorter-term horizons, and PE/VC funds rely on liquid stock markets to exit their positions. Transparent, resilient, and liquid capital markets are also needed to attract long-term institutional investors capable of taking on bigger risks in alternative or non-traditional financial instruments. Mobilizing insurance, pension, and social-security funds – which have the long-term savings and investment horizons to fund technological innovation (as well as infrastructure investments) – is essential.

So far, China's leaders have struggled to build the innovation ecosystem the country needs, including a liquid and transparent public and private funding mechanism. In the context of the Sino-American rivalry, this amounts to a massive disadvantage. To compete with the US, China needs to foster its own tech unicorns and build global platforms that match their American counterparts in speed, scale, and scope.

The lead is secure — their studies routinely misunderstand the US market economy.

Lewis 24 — James Andrew Lewis, Senior Vice President, Pritzker Chai, Director of the Strategic Technologies Program, 2024 (“America is Losing the Shoe Race With China,” *CSIS*, April 1st, Available Online at <https://www.csis.org/blogs/working-papers/america-losing-shoe-race-china>, Accessed 09-04-2024)

A better question might be who cares. The number of shoes is not a good indicator of national power. In fact, no single technology is a good indicator of national power. The U.S. economy is vast, decentralized, continental in size, and is guided by actively competitive markets. It has been exceptionally innovative for decades. Leading in a single technology (like railroads in the 19th century or semiconductors today) reflects a common analytical error that misjudges how economies and technology actually create national power. The concept of a “race” itself is a questionable legacy of Cold War thinking – the Cold War had a finish line (identified by Eisenhower and Dulles at the onset), while the current situation does not.

Stories about the United States falling behind are so predictable that they form a literary genre. In 1957, the President's Science Advisor predicted that Soviet performance in math and science education would

give it global leadership in a decade. In 1969, the Departments of Treasury, Commerce and Agriculture warned President Nixon that a powerful new economic entity, the European Union, would displace the United States. Starting the 1980s, assorted pundits announced that Japan would dominate the global economy. And until recently, there were routine predictions that China would displace the United States, predictions that still make regular appearances.

These predictions have two things in common. First, they were wrong. Second, they were wrong because they counted the wrong things. They did not place their analyses in the context of larger national economies. Instead, they relied on picking illustrative metrics, usually proxy indicators that provide an indirect measurement of technological success. One recurring problem is the tendency to measure inputs rather than outcomes. A politician may have great staff and spend more on an election, but the ultimate metric is how many votes are received. Claims that the United States is falling behind China in 5G because China has deployed more base stations or has more 5G enabled phones reflects a similar confusion over metrics. It is not the number of base stations that is important, it is the ability to use 5G to create new goods and services (or be more efficient in the use of existing goods and services) that is important, and this is best measured by the monetization of new 5G enabled services and products and their revenue.

A report that announces that China leads in 37 technologies out of 44 technologies does not explain why the United States is the center for development of artificial intelligence, quantum technology, and biotechnology. China did not develop successful COVID vaccines, lags in quantum, and there are anecdotes that China's leaders asked the author of a best-selling 2018 book on China's coming dominance of artificial intelligence why, if that was the case, were GPT technologies developed first in the United States? Of the 37 technologies listed, China's alleged lead is open to question in 23. Does China really lead in cybersecurity, as the Report asserts? The digital economy depends on cloud computing, next generation networks (like 5G and 6G), and software (like AI products) and these are technologies where the United States has a strong if not dominant position.

Computing

Cyber Turn

Turn - Increased patent protections decrease competition which decks cybersecurity broadly.

Duan 20 — Charles Duan, Director of Technology and Innovation Policy, R Street Institute, Washington, D.C., 2022 ("Of Monopolies and Monocultures: The Intersection of Patents and National Security," *Santa Clara High Technology Law Journal*, Vol. 36, no. 4, Article 4, May 11th, Available Online at <https://digitalcommons.law.scu.edu/chtlj/vol36/iss4/5/>, Accessed 08-23-2024)

This syllogistic equating of patent protection and national security, though perhaps initially attractive, appears on closer inspection to be overly simplistic. The purpose of this Article is to challenge this line of reasoning, showing that patent rights, and in particular unbridled assertion of patents, can impair and repeatedly has impaired national security interests. It reaches this conclusion by considering patents and

national security through the lens of competition. Patents by definition suppress competition to some degree,^{1 2} so insofar as competition can enhance national security in certain ways, patents can diminish national security when used or licensed in particularly aggressive ways. To establish the relationship between patents and national security, then, this Article also contemplates how competition relates to national security.

The Article proceeds as follows. Part II reviews how patent policy has recently intersected with national security. It first explains the primary basis for relating innovation policy generally with national security, namely ongoing technological "races" in security-sensitive technologies. It then discusses recent events that have given rise to interest in and arguments over how patent law and policy can affect national security interests, particularly those relating to the aforementioned technology races, and identifies the appearance of the syllogistic argument that patent protection uniformly increases national security.

The remainder of the Article interrogates the syllogistic argument, primarily on two fronts. Part II takes a historical perspective, considering three past anecdotes in which patents have run headlong into national security.³ While each of the three examples—torpedo development prior to World War I, patent licensing in early aviation, and bioterrorism threats following the September 11th attacks—offers unique insights and lessons (plus an unexpected factoid on *The Sound of Music* ⁴), the common thread is that aggressive assertion and licensing of patents, by creating an environment devoid of competition, can stymie important government interests in national defense and security. Part III considers cybersecurity, which is closely tied to national security.¹⁵ Reviewing economic and computer science research, Part III identifies two reasons why competition enhances cybersecurity: Competition encourages firms to improve cybersecurity as a market-driven value-add, and it prevents the formation of single-vendor "monocultures" of technology that have been shown to be especially susceptible to cyberattacks.

Cyberattacks cause global war — that's their Perkovich.

Cyberattacks Escalate — exacerbates NC3 vulnerabilities which trips deterrence.

Klare 19 — Michael T. Klare, Professor emeritus of peace and world security studies at Hampshire College and senior visiting fellow at the Arms Control Association, 2019 ("Cyber Battles, Nuclear Outcomes? Dangerous New Pathways to Escalation," *Arms Control Association*, November, Available Online at <https://www.armscontrol.org/act/2019-11/features/cyber-battles-nuclear-outcomes-dangerous-new-pathways-escalation>, Accessed 08-23-2024)

Another initiative incorporated in the strategy document also aroused concern: the claim that an enemy cyberattack on U.S. nuclear command, control, and communications (NC3) facilities would constitute a "non-nuclear strategic attack" of sufficient magnitude to justify the use of nuclear weapons in response.

Under the Obama administration's NPR report, released in April 2010, the circumstances under which the United States would consider responding to non-nuclear attacks with nuclear weapons were said to be few. "The United States will continue to...reduce the role of nuclear weapons in deterring non-nuclear attacks," the report stated. Although little was said about what sort of non-nuclear attacks might be deemed severe enough to justify a nuclear response, cyberstrikes were not identified as one of these. The 2018 NPR report, however, portrayed a very different environment, one in which nuclear combat is seen as increasingly possible and in which non-nuclear strategic threats, especially in cyberspace, were viewed as sufficiently menacing to justify a nuclear response. Speaking of Russian technological progress, for example, the draft version of the Trump administration's NPR report stated, "To...correct any Russian misperceptions of advantage, the president will have an expanding range of limited and graduated [nuclear] options to credibly deter Russian nuclear or non-nuclear strategic attacks, which could now include attacks against U.S. NC3, in space and cyberspace."¹

The notion that a cyberattack on U.S. digital systems, even those used for nuclear weapons, would constitute sufficient grounds to launch a nuclear attack was seen by many observers as a dangerous shift in policy, greatly increasing the risk of accidental or inadvertent nuclear escalation in a crisis. "The entire broadening of the landscape for nuclear deterrence is a very fundamental step in the wrong direction," said former Secretary of Energy Ernest Moniz. "I think the idea of nuclear deterrence of cyberattacks, broadly, certainly does not make any sense."²

Despite such admonitions, the Pentagon reaffirmed its views on the links between cyberattacks and nuclear weapons use when it released the final version of the NPR report in February 2018. The official text now states that the president must possess a spectrum of nuclear weapons with which to respond to “attacks against U.S. NC3,” and it identifies cyberattacks as one form of non-nuclear strategic warfare that could trigger a nuclear response.

That cyberwarfare had risen to this level of threat, the 2018 NPR report indicated, was a product of the enhanced cybercapabilities of potential adversaries and of the creeping obsolescence of many existing U.S. NC3 systems. To overcome these vulnerabilities, it called for substantial investment in an upgraded NC3 infrastructure. Not mentioned, however, were extensive U.S. efforts to employ cybertools to infiltrate and potentially incapacitate the NC3 systems of likely adversaries, including Russia, China, and North Korea.

For the past several years, the U.S. Department of Defense has been exploring how it could employ its own very robust cyberattack capabilities to compromise or destroy enemy missiles from such states as North Korea before they can be fired, a strategy sometimes called “left of launch.”³ Russia and China can assume, on this basis, that their own launch facilities are being probed for such vulnerabilities, presumably leading them to adopt escalatory policies such as those espoused in the 2018 NPR report. Wherever one looks, therefore, the links between cyberwar and nuclear war are growing.

The Nuclear-Cyber Connection

These links exist because the NC3 systems of the United States and other nuclear-armed states are heavily dependent on computers and other digital processors for virtually every aspect of their operation and because those systems are highly vulnerable to cyberattack. Every nuclear force is composed, most basically, of weapons, early-warning radars, launch facilities, and the top officials, usually presidents or prime ministers, empowered to initiate a nuclear exchange. Connecting them all, however, is an extended network of communications and data-processing systems, all reliant on cyberspace. Warning systems, ground- and space-based, must constantly watch for and analyze possible enemy missile launches. Data on actual threats must rapidly be communicated to decision-makers, who must then weigh possible responses and communicate chosen outcomes to launch facilities, which in turn must provide attack vectors to delivery systems. All of this involves operations in cyberspace, and it is in this domain that great power rivals seek vulnerabilities to exploit in a constant struggle for advantage.

The use of cyberspace to gain an advantage over adversaries takes many forms and is not always aimed at nuclear systems. China has been accused of engaging in widespread cyberespionage to steal technical secrets from U.S. firms for economic and military advantages. Russia has been accused, most extensively in the Robert Mueller report, of exploiting cyberspace to interfere in the 2016 U.S. presidential election. Nonstate actors, including terrorist groups such as al Qaeda and the Islamic State group, have used the internet for recruiting combatants and spreading fear. Criminal groups, including some thought to be allied with state actors, such as North Korea, have used cyberspace to extort money from banks, municipalities, and individuals.⁴ Attacks such as these occupy most of the time and attention of civilian and military cybersecurity organizations that attempt to thwart such attacks. Yet for those who worry about strategic stability and the risks of nuclear escalation, it is the threat of cyberattacks on NC3 systems that provokes the greatest concern.

This concern stems from the fact that, despite the immense effort devoted to protecting NC3 systems from cyberattack, no enterprise that relies so extensively on computers and cyberspace can be made 100 percent invulnerable to attack. This is so because such systems employ many devices and operating systems of various origins and vintages, most incorporating numerous software updates and “patches” over time, offering multiple vectors for attack. Electronic components can also be modified by hostile actors during production, transit, or insertion; and the whole system itself is dependent to a considerable degree on the electrical grid, which itself is vulnerable to cyberattack and is far less protected. Experienced “cyberwarriors” of every major power have been working for years to probe for weaknesses in these systems and in many cases have devised cyberweapons, typically, malicious software (malware) and computer viruses, to exploit those weaknesses for military advantage.⁵ Although activity in cyberspace is much more difficult to detect and track than conventional military operations, enough information has become public to indicate that the major nuclear powers, notably China, Russia, and the United States, along with such secondary powers as Iran and North Korea, have established extensive cyberwarfare capabilities and engage in offensive cyberoperations on a regular basis, often aimed at critical military infrastructure. “Cyberspace is a contested environment where we are in constant contact with adversaries,” General Paul M. Nakasone, commander of the U.S. Cyber Command (Cybercom), told the Senate Armed Services Committee in February 2019. “We see near-peer competitors [China and Russia] conducting sustained campaigns below the level of armed conflict to erode American strength and gain strategic advantage.”

Although eager to speak of adversary threats to U.S. interests, Nakasone was noticeably but not surprisingly reluctant to say much about U.S. offensive operations in cyberspace. He acknowledged, however, that Cybercom took such action to disrupt possible Russian interference in the 2018 midterm elections. “We created a persistent presence in cyberspace to monitor adversary actions and crafted tools and tactics to frustrate their efforts,” he testified in February. According to press accounts, this included a cyberattack aimed at paralyzing the Internet Research Agency, a “troll farm” in St. Petersburg said to have been deeply involved in generating disruptive propaganda during the 2016 presidential elections.⁶

Other press investigations have disclosed two other offensive operations undertaken by the United States. One called “Olympic Games” was intended to disrupt Iran’s drive to increase its uranium-enrichment capacity by sabotaging the centrifuges used in the process by infecting them with the so-called Stuxnet virus. Another left of launch effort was intended to cause malfunctions in North Korean missile tests.⁷ Although not aimed at either of the U.S. principal nuclear adversaries, those two attacks demonstrated a willingness and capacity to conduct cyberattacks on the nuclear infrastructure of other states. Efforts by strategic rivals of the United States to infiltrate and eventually degrade U.S. nuclear infrastructure are far less documented but thought to be no less prevalent. Russia, for example, is believed to have planted malware in the U.S. electrical utility grid, possibly with the intent of cutting off the flow of electricity to critical NC3 facilities in the event of a major crisis.⁸ Indeed, every major power, including the United States, is believed to have crafted cyberweapons aimed at critical NC3 components and to have implanted malware in enemy systems for potential use in some future confrontation.

Pathways to Escalation

Knowing that the NC3 systems of the major powers are constantly being probed for weaknesses and probably infested with malware designed to be activated in a crisis, what does this say about the risks of escalation from a nonkinetic battle, that is, one fought without traditional weaponry, to a kinetic one, at first using conventional weapons and then, potentially, nuclear ones? None of this can be predicted in advance, but those analysts who have studied the subject worry about the emergence of dangerous new pathways for escalation. Indeed, several such scenarios have been identified.⁹

The first and possibly most dangerous path to escalation would arise from the early use of cyberweapons in a great power crisis to paralyze the vital command, control, and communications capabilities of an adversary, many of which serve nuclear and conventional forces. In the “fog of war” that would naturally ensue from such an encounter, the recipient of such an attack might fear more punishing follow-up kinetic attacks, possibly including the use of nuclear weapons, and, fearing the loss of its own arsenal, launch its weapons immediately. This might occur, for example, in a confrontation between NATO and Russian forces in east and central Europe or between U.S. and Chinese forces in the Asia-Pacific region.

Speaking of a possible confrontation in Europe, for example, James N. Miller Jr. and Richard Fontaine wrote that “both sides would have overwhelming incentives to go early with offensive cyber and counter-space capabilities to negate the other side’s military capabilities or advantages.” If these early attacks succeeded, “it could result in huge military and coercive advantage for the attacker.” This might induce the recipient of such attacks to back down, affording its rival a major victory at very low cost. Alternatively, however, the recipient might view the attacks on its critical command, control, and communications infrastructure as the prelude to a full-scale attack aimed at neutralizing its nuclear capabilities and choose to strike first. “It is worth considering,” Miller and Fontaine concluded, “how even a very limited attack or incident could set both sides on a slippery slope to rapid escalation.”¹⁰

What makes the insertion of latent malware in an adversary’s NC3 systems so dangerous is that it may not even need to be activated to increase the risk of nuclear escalation. If a nuclear-armed state comes to believe that its critical systems are infested with enemy malware, its leaders might not trust the information provided by its early-warning systems in a crisis and might misconstrue the nature of an enemy attack, leading them to overreact and possibly launch their nuclear weapons out of fear they are at risk of a preemptive strike.

“The uncertainty caused by the unique character of a cyber threat could jeopardize the credibility of the nuclear deterrent and undermine strategic stability in ways that advances in nuclear and conventional weapons do not,” Page O. Stoutland and Samantha Pitts-Kiefer wrote in 2018 paper for the Nuclear Threat Initiative. “[T]he introduction of a flaw or malicious code into nuclear weapons through the supply chain that compromises the effectiveness of those weapons could lead to a lack of confidence in the nuclear deterrent,” undermining strategic stability.¹¹ Without confidence in the reliability of its nuclear weapons infrastructure, a nuclear-armed state may misinterpret confusing signals from its early-warning systems and, fearing the worst, launch its own nuclear weapons rather than lose them to an enemy’s first strike. This makes the scenario proffered in the 2018 NPR report, of a nuclear response to an enemy cyberattack, that much more alarming.

Patents create technological monocultures that collapse cybersecurity.

Duan 20 — Charles Duan, Director of Technology and Innovation Policy, R Street Institute, Washington, D.C., 2022 (“Of Monopolies and Monocultures: The Intersection of Patents and National Security,” *Santa*

In addition to the historical review done so far, another approach to understanding the relationship among patents, competition, and national security is to consider the role of cybersecurity. There is little doubt that computer system vulnerabilities that enable hacking and spread of computer exploits are a threat to the nation's defenses, so better cybersecurity is a key part of national security strategy. 155

Strong competition can thus complement national security by enhancing domestic cybersecurity, and patent assertion that unduly weakens competition [*395] detracts from cybersecurity. 156 Competition promotes better cybersecurity in at least two ways. First, multiple studies show that competition encourages firms to improve their products on multiple vectors including cybersecurity. Second, competition avoids a situation that security experts call a "monoculture," which increases vulnerability to severe cyberattacks. As former Secretary of Homeland Security Michael Chertoff wrote recently, "We need competition and multiple providers, not a potentially vulnerable technological monoculture, to guarantee national security". 157 Thus, cybersecurity provides a useful lens for understanding how unfettered patent assertion and licensing can detract from national security.

A. Cybersecurity as Competitive Value-Add

Competition enhances national security by reducing the incidence of technical vulnerabilities. That effect is especially important for security-sensitive systems such as mobile telecommunications.

Intuitively, a causal chain from competition to cybersecurity makes logical sense. Computer security is a value-added benefit to consumers, so firms in competitive markets are likely to use security to gain an edge over their competitors. 158 In monopolized markets, though, there may be less external impetus to test products for flaws, and the monopolist may choose to focus less on security and more on new product features or increased product quality.

Economic research confirms these hypotheses about competition leading to better cybersecurity. A 2009 empirical study of web browsers considered the impact of market concentration on the amount of time that vendors took to fix security vulnerabilities as they were discovered. 159 The study found that the presence of more competitors correlated with faster cybersecurity response--a reduction of 8-10 days in response time per additional market rival. 160 Similarly, business researchers in 2005 modeled incentives for firms to engage in sharing of cybersecurity information, and concluded that the "inclination to share information and invest in security technologies increases" [*396] as the degree of competitiveness in an industry increases. 161 Another study found that, where two software firms are in competition, at least one will be willing to take on some degree of risk and responsibility for cybersecurity, whereas a monopoly software firm will consistently fail to accept such responsibility. 162 To be sure, an unpublished study from 2017 found that some market concentration can make firms more responsive to cybersecurity issues, but only to a point: "being in a dominant position reduces the positive effect of having less competitors on the responsiveness of the vendor," and indeed the "more dominant the firm is, the less rapid it is in releasing security patches." 163 This research confirms that competition is more conducive to cybersecurity.

It is not hard to see how this applies to emerging communication technologies markets. In the absence of competition, the above research suggests that device manufacturers, chip makers, and software developers will lack incentives to respond to vulnerabilities, to share information about cybersecurity practices and issues, and to take responsibility for security matters. Mobile phone chips have had their share of cybersecurity

failures already. 164 The best way to flush out ongoing and future cybersecurity issues is to maintain competitive pressure at all levels of the supply chain.

B. Vulnerabilities of "Monocultures"

A second reason why monopoly undermines cybersecurity is that monopoly leads to a "monoculture" of single-vendor products, opening the door to massive systemic failure in the case of a cyberattack.

Computer researchers developed the theory of software monocultures in the early 2000s, in response to the regular phenomenon of computer viruses and other attacks spreading rapidly by exploiting flaws in the dominant operating system at the time, Microsoft Windows. 165 Where a computer system such as Windows has a commanding share of users, a virus that exploits a flaw in that system can [*397] quickly spread to infect a whole interconnected ecosystem. An operating system monopoly thus enables fast and easy spread of cyberattacks, and better cybersecurity would be achieved through greater diversity in online systems. 166As one research group posited, "a network architecture that supports a collection of heterogeneous network elements for the same functional capability offers a greater possibility of surviving security attacks as compared to homogeneous networks." 167

There has been considerable study of the theory that computer monocultures are naturally more vulnerable to attacks. 168 In one study, computer science researchers reviewed a catalog of 6,340 software vulnerabilities recorded in 2007, to compare whether comparable software would share the same flaws. 169 Of the 2,627 vulnerabilities applicable to application software (as opposed to operating systems, web scripts, and other software components), only 29 (1.1%) applied to substitute products from different vendors but providing the same functionality. 170By contrast, different versions of a single software product were found to share vulnerabilities 84.7% of the time. 171Thus, software monocultures share exploitable flaws even when there is some variation in versions across the monoculture; by contrast, diversity in software is almost guaranteed to prevent a single flaw from affecting all users.

In the case of 5G and wireless mobile communications, a monoculture is an especially concerning possibility. To the extent that systems such as smart city sensors or communication networks are widely deployed in a monoculture fashion, a widespread attack could have devastating consequences, potentially blacking out a region and affecting essential services such as 911. 172 A monoculture that is vulnerable to so-called "rootkits" or "backdoors"--maliciously installed software that enable bad actors to commandeer systems--could also enable mass surveillance or spying by private hackers or foreign governments. 173The presence of systems from multiple vendors would mitigate these possibilities.

[*398] The monoculture theory is not without critics, but a review of those criticisms shows them to be inapplicable to contemporary communication technologies. Some critics suggest that software diversity imposes unwarranted costs on firms who must forego economies of scale and devise seemingly duplicative yet different setups of computer systems. 174 But those concerns largely focus on the situation where a single firm produces and manages heterogeneous systems, concerns that are avoided where heterogeneity arises naturally through competition between two unrelated firms. Critics also argue that technological measures can create "artificial diversity" through automated randomization of software code, so software engineers can purportedly solve monoculture issues and device users need not worry about the issue. 175But even these critics acknowledge that artificial diversity techniques are often insufficient because they must make assumptions about what aspects of the technology are most vulnerable to attack, and they concede that artificial diversity cannot stop attacks involving operation of legitimate software functions in undesirable ways (sending spam emails or deleting document files, for example). 176

Patents don't incentivize computer innovation and actually serve to prevent innovation in the software industry.

Abbott 16 — Ryan Abbott, Doctor of Philosophy from the University of Surrey, MD from the University of California, San Diego, JD from Yale Law School, MTOM, Summa Cum Laude from Emperor's College, BS, Summa Cum Laude from the University of California, Las Angeles, Professor and Chair of Law and Health Sciences at the University of Surrey, Director of Research at the University of Surrey Law School,

School of Law Researcher of the Year, University of Surrey "Innovator of the Year," 2016, ("I Think, Therefore I Invent: Creative Computers and the Future of Patent Law," *Boston College Law*, September 28th, Available Online at <https://lira.bc.edu/work/sc/39472700-d7fc-4c23-9b47-a7f631296a2e>, Accessed 08-27-2024)

Those arguments reflect the dominant narrative justifying the grant of intellectual property protection.¹⁸⁹ That account, however, has been criticized, particularly by academics.¹⁹⁰ Patents result in significant social costs by establishing monopolies.¹⁹¹ Patents also can stifle entry by new ventures by creating barriers to subsequent research.¹⁹² Whether the benefit of patents as an innovation incentive outweighs their anti-competitive costs, or for that matter, whether patents even have a net positive effect on innovation, likely varies between industries, areas of scientific research, and inventive entities.

For instance, commentators such as Judge Richard Posner have argued that patents may not be needed to incentivize R&D in the software industry.¹⁹⁴ Software innovation is often relatively inexpensive, incremental, quickly superseded, produced without patent incentives, protected by other forms of intellectual property, and associated with a significant first mover advantage.¹⁹⁵ Likewise, patents may be unnecessary to spur innovation in university settings where inventors are motivated to publish their results for prestige and the prospect of academic advancement.

Computational inventions may develop due to non-patent incentives. Software developers have all sorts of non-economic motivations to build creative computers: for example, to enhance their reputations, satisfy scientific curiosity, or collaborate with peers.¹⁹⁷ Business ventures might find the value of computational inventions exceeds the cost of developing creative computers even in the absence of patent protection. Of course, computational invention patents may not be an all-or-nothing proposition; they may further encourage activities that would have otherwise occurred on a smaller scale over a longer timeframe. If patents are not needed to incentivize the development of creative computers, it may be justifiable to treat computational inventions as unpatentable and failing to recognize computer inventors. Yet, whether patents produce a net benefit as an empirical matter is difficult to determine a priori. Even though individuals and businesses do not always behave as rational economic actors, in the aggregate, it is likely that providing additional financial incentives to spur the development of creative computers will produce a net benefit.

Patents for computational inventions might also be opposed on the grounds that they would chill future human innovation, reward human inventors who failed to contribute to the inventive process, and result in further consolidation of intellectual property in the hands of big business (assuming that businesses such as IBM will be the most likely to own creative computers).¹⁹⁹

Other non-utilitarian patent policies do not appear to support computer inventorship. For example, courts have justified granting patent monopolies on the basis of Labor Theory, which holds that a person has a natural right to the fruits of their work.²⁰⁰ Labor Theory may support giving a patent to someone who has worked for years to invent a new device so that they can profit from their invention, but it does not apply to computers because computers cannot own property. All computer work is appropriated. Similarly, Personality Theory, which holds that innovation is performed to fulfill a human need, would not apply to AI.²⁰¹ Creative computers invent because they are instructed to invent, and a machine would not be offended by the manner in which its inventions were used. AI might even be a concerning recipient for inventorship under Social Planning Theory, which holds that patent rights should be utilized to promote cultural goals.²⁰² An AI could develop immoral new technologies.²⁰³ Submissions, however, are no longer rejected by the Patent Office for being "deceitful" or "immoral," and, to the extent this is a concern, there would be opportunities for a person to judge the morality of an application before it is granted.

Patents kill quantum computing innovation — creates roadblocks.

Kop et al 22 — Mauritz Kop, Founder and Executive Director of the Stanford Center for Responsible Quantum Technology, Fellow at Stanford University, Fellow at the Novo Nordisk Foundation Copenhagen-Cambridge-Harvard Inter-CeBIL Programme for International Collaborative Bioscience Innovation & Law, Founder of MusicaJuridica, Director at AIRecht, and General Counsel at Daiki, member of the European AI Alliance, M.A. in Law from Maastricht University, Mateo Aboy, Director of Research in Technology & Law at the University of Cambridge, Professor & Fellow at the University of Copenhagen, Chairman of the Board of Directors at APDM Wearable Technologies, PhD Doctor of Juridical Science & Doctor of Laws from the University of Salamanca/LML-University of Cambridge & PFC-Harvard, PhD in Electrical, Computer & Communications Engineering from the University of Vigo, MS in AI/Machine Learning, DSP & Quantum Information Science from the Universidade de Vigo, Law Degree from the University of London, MBA in International Business Management from the University of London, Timo Minssen, Professor of Law at the University of Copenhagen, Founding Director of UCPH's Center for Advanced Studies in Bioscience Innovation Law, German law degree from Georg-August-University in Göttingen, LL.M. degree from Lund University, 2022 ("Intellectual property in quantum computing and market power: a theoretical discussion and empirical analysis," *The Oxford Academic Journal of Intellectual Property Law & Practice*, July 19th, Available Online at <https://academic.oup.com/jiplp/article/17/8/613/6646536>, Accessed 09-12-2024)

However, overprotection through IPRs can also lead to a situation that would create a fundamentally anticompetitive environment.⁶ For example, a proliferation of patent rights upstream could potentially hinder essential innovations further downstream in the course of scientific research and product development because each upstream patent allows its owner to create another obstacle on the road to product development, adding to the cost and slowing the pace of downstream innovation.⁷

Dealing with this potential dilemma, commonly referred to as the 'Tragedy of the Anticommons', requires a reasonable balance to be found between the stimulation of innovation competition, the enhancement of scientific research and the careful protection of intellectual property rights.⁸ To maintain such a reasonable balance, each new technology has involved modifications to the law. This is nothing new. The first patents, during the Industrial Revolution, were mostly directed to mechanical devices and articles of manufacture. When standardized chemical nomenclature developed, case law refined the existing legal framework to solve new problems posed by compositions and classifications of matter.⁹ Information age inventions based on computer technology, synthetic biotechnology and gene editing, fuelled by the evolution of integrated circuits (ICs) and software, as well as stunning discoveries in molecular biology and nanotechnology have also led to many legal debates and a refinement of case law and legislation.

We are currently on the verge of a new technological revolution associated with 2G quantum technologies, including quantum computing. Some commentators anticipate its complexity might create challenges, which could never have been foreseen when the IPR system was developed.¹⁰ Against this background, this article (a) inquires which IPRs and IP-related rights would primarily be applicable to quantum technology and (b) provides an analysis of whether the strategic use of integrated mixtures of IP rights to maximize the value of a quantum computer owner's IP portfolio potentially might lead to anticompetitive situations. Overprotection of information causes market barriers and hinders both healthy competition and industry-specific innovation. In this case, it would slow down progress in an important application area of quantum technology, namely quantum computing. Particular emphasis will

therefore be laid on the question if, and if so to what degree, overlapping IPRs could result in an inappropriate duration of global exclusive exploitation rights for first movers and in an unwanted concentration of market power.